LABORATOIRE HUBERT CURIEN
UMR • CNRS • 5516 • SAINT-ETIENNE

Fouille de textes
Text-Mining

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Who am I ? C. LARGERON

Professor of Computer Science, Université Jean Monnet
Adjunct professor, University of Alberta - Canada

What am I doing:
Head of Master DSMI - School of Economy SE2

Research in Data mining and Machine learning
Text mining, social mining
Information retrieval
Hubert Curien Laboratory - UMR CNRS 5416
How to join me :
04/77/91/57/56
Christine.Largeron@univ-st-etienne.fr

What are we doing together ?

Text and social mining
Lessons 1 & 2: text mining (12 sept L2010 – 14 sept – L216
Lessons 3 & 4: social mining (16 nov – 23 Nov – L216)
Practical Lesson : R Vaudaine

Evaluation: Exercices & examination & project

Examination: Exercices based on the course (C.Largeron + B Jeudy)

Project: see later

Prerequisites

Classical data mining methods:

- Principal composant analysis (PCA)
- Clustering methods:
  - Partitioning method ie K-means
  - Hierarchical algorithms (single/ complete linkage)
- Supervised clusering (Decision trees, SVM, etc.)

References

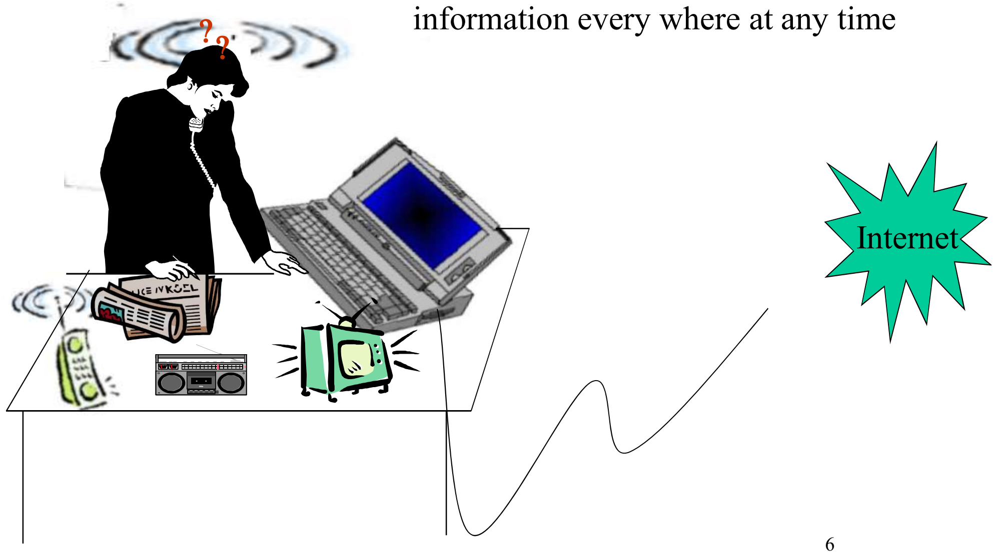
- Gilbert Saporta « Probabilités, analyse des données et Statistique » Edition Technip. (Chap 8 – 12) https://www.fichier-pdf.fr/2012/06/15/probabilites-analyse-de-donnees-et-statistiques/
- Han J, Kamber M.(2001) Data mining: Concepts and techniques, Morgan Kaufman (Chap 7 – 8)
- Mitchell.T. Machine Learning. McGraw-Hill, 1997 (Chap 3)
- Cornuejols A and Miclet L Apprentissage artificiel : Concept et algorithmes Eyrolles 2002 (Chap 11 – 15)

Outline

- Introduction
- Text preprocessing
- Features extraction - indexing
- Weighting models
- Document similarity
- Features Extraction - dimension reduction
- Text mining (categorization – clustering- association)
- Application with R

Introduction : What happens

• Rapid proliferation of information available in digital format
•Current technologies allow to obtain this information every where at any time



Introduction : What happens

• 80% information stored in text documents: journals, web pages, emails...

•90% of web data is text
http://www.emc.com/leadership/digital-universe/index.htm

• Most information is free text, not in structured data

• People have less time to access the data and absorb information

Introduction : What happens

It is necessary to automatically analyze,organize,summarize... all this textual data

Introduction : What is text mining

Definition : Data Mining

« The non trivial extraction of implicit, <u>previously unknown</u>, and potentially useful information from data » [1]

« The science of extracting useful information from large data sets or databases » [2]

[1] W. Frawley and G. Piatetsky-Shapiro and C. Matheus (Fall 1992). "Knowledge Discovery in Databases: An Overview". AI Magazine: pp. 213-228. ISSN 0738-4602.

[2] D. Hand, H. Mannila, P. Smyth (2001). Principles of Data Mining. MIT Press, Cambridge, MA. ISBN 0-262-08290-X.

Introduction : What is text mining

Definition : Data Mining

« The non trivial extraction of implicit, <u>previously unknown</u>, and potentially useful information from data »

<> Querying a database = Extracting useful patterns

Select ID from Table where Internet = Yes;

Predictive model (decision tree, SVM, etc) , clusters, association rules, …

ID	Age	Genre	Internet
1	12	M	No
2	27	F	Yes
3	84	F	Yes

Introduction : What is text mining

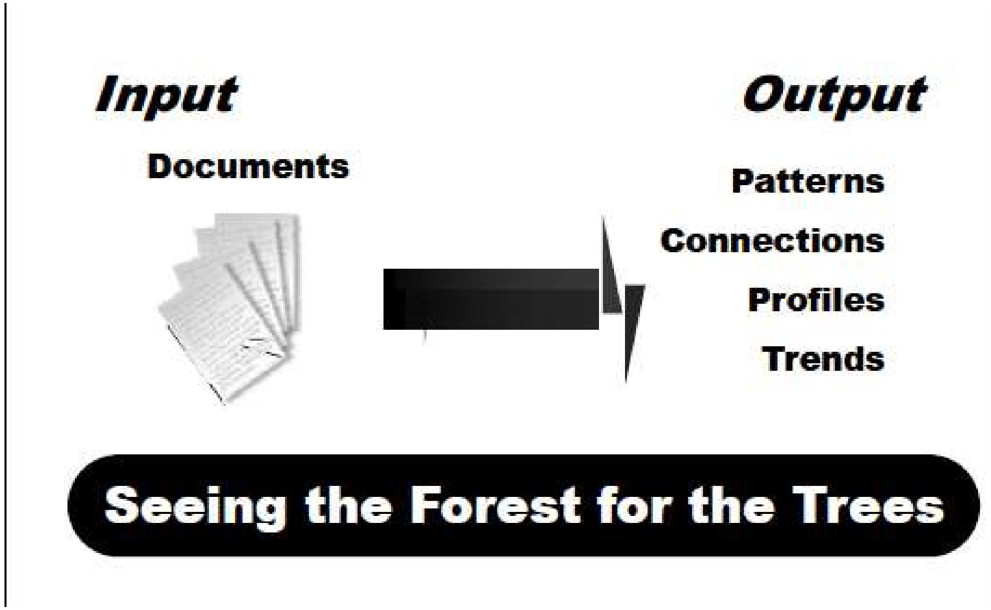
Definition : Text Mining

"The non trivial extraction of implicit, previously unknown, and potentially useful information from (large amount of) textual data".

=>

" An exploration and analysis of textual (natural language) data by automatic and semi automatic means to discover new knowledge".

Introduction : What is TM on document databases?



Introduction :What can text mining do ?

- The potential applications are countless.
  - Customer profile analysis (for advertising, etc.)
  - Mel / Information filtering and routing
  - News/ stories/web pages classification
  - Trend analysis
  - Event tracks
  - Web search
  - …….

Introduction : What is not text mining

Text mining is not Information Retrieval (IR)

Given

- a collection of documents
- a user need (example documents, keywords, query given in a search engine like google or exalead or bing…)

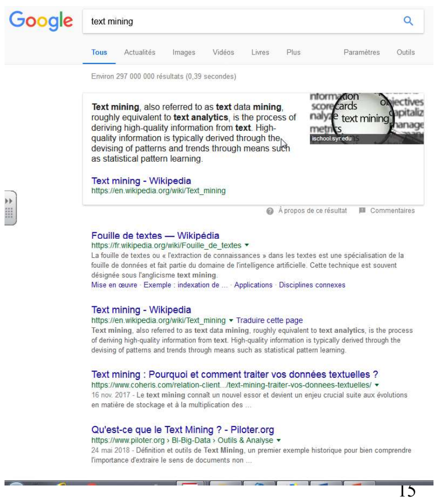
retrieve relevant documents in the collection

Introduction : What is not text mining

Text mining is not Information Retrieval (IR)

retrieve relevant documents in the collection

TREC (Text Retrieval Conference)
http://trec.nist.gov/
INEX
https://inex.mmci.uni-saarland.de/
ACM SIGIR: Information Retrieval Special Interest Group



Introduction : What is not text mining

Text mining is not Information Extraction

Extracting structured information from unstructured and/or semi-structured machine-readable documents.

Example Housing offers

CENT.VILLE SAINT ETIENNE Rue Henri Gonard, F3 en duplex 70m2 1ét.,chauf.elct,bon.isol.,1.860 €. Tél.04.77.57.09.22

⇒ *town*: CENT.VILLE SAINT ETIENNE
- *address* = Rue Henri Gonard
- *type* = **F3**
- *area* = 70 **m²**
- *rent* = 1.860 ***€***
- *Contact* = ***Tél****.04.77.57.09.22*

Message Understanding Conferences (MUC)

Introduction : What is not text mining

Text mining is not Natural Language Processing

- •Morphological / lexical segmentation of <u>one text</u> in units : words, sentences
- •Syntactic analysis (combination of words in the sentence)

  Jean (nom) mange (verbe) des (déterminants) pommes de terre (nom)
  Jean (noun) eats (verb) potatoes (noun)
- •Semantic analysis (meaning of the words out of the context)
- •Pragmatic analysis (meaning of the words in the context)

  La mousse aux fraises est sur la table de l'avocat (fruit or layer).
  The lean man stopped to lean against the post (lean: adjective or verb)
  Sherlock saw the man with binoculars

Automatic summarization
Machine translation
Named entity recognition
Relationship extraction

Introduction : Text sources/document types

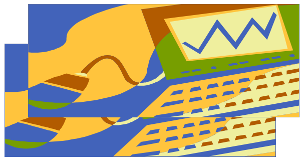
Text sources

- Comments and notes
  - Physicians, Sales reps.
  - Customer response centers
  - Email
  - Word & PowerPoint documents
- The web
  - blogs
- Journal articles
  - Medline has 13 million abstracts
- Annotations in databases
  - e.g. GenBank, GO, EC, PDB

Document types

- Structured documents
  - Output from CGI
- Semi-structured documents
  - Seminar announcements
  - Job listings
  - Ads
- Free format documents
  - News
  - Scientific papers
  - Blogs

Introduction :Document types



Structured documents
Ads

HomeLoan (
Loanee: *Frank Rizzo*
Lender: *MWF*
Agency: *Lake View*
Amount: *$200,000*
Term: *15 years*
)

Multimédia
News paper

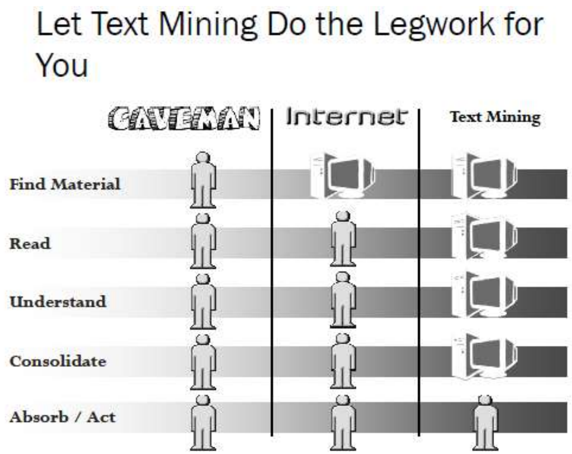
Free format
Book

Frank Rizzo bought his home from Lake View Real Estate in 1992.
He paid $200,000 under a15-year loan from MW Financial.

Semi structured
Web page
(html – XML – scientific paper)

<a href>*Frank Rizzo* *Bought* <a hef>*this home* *from* <a href>*Lake View Real Estate* *In* *1992*. <p>...

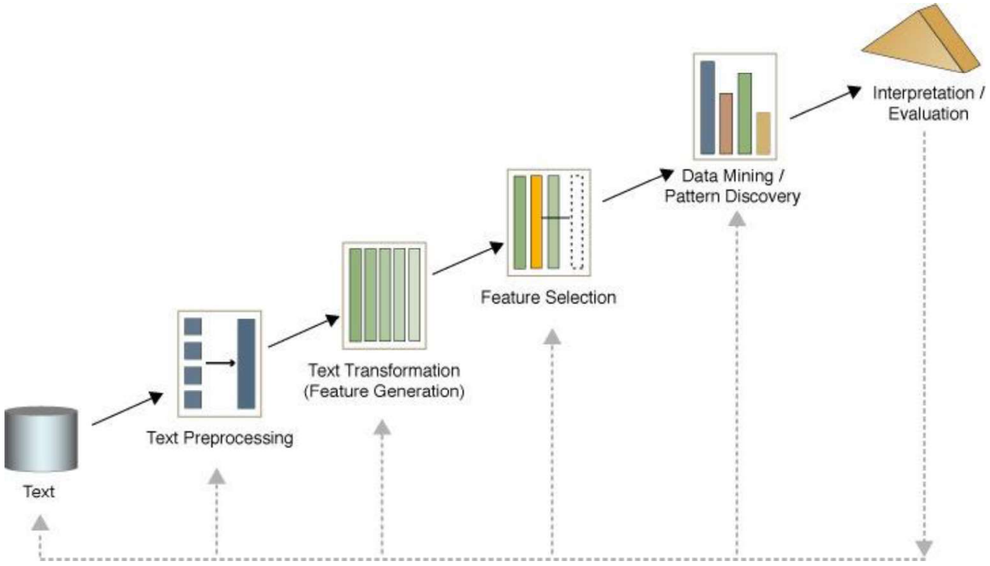
Introduction : How does it work ?



Text mining: automatization of the main parts of the task in such a way that the user has only to analyse the knowledge extracted from the data

Introduction : How does it work ?

…like data mining but on textual databases

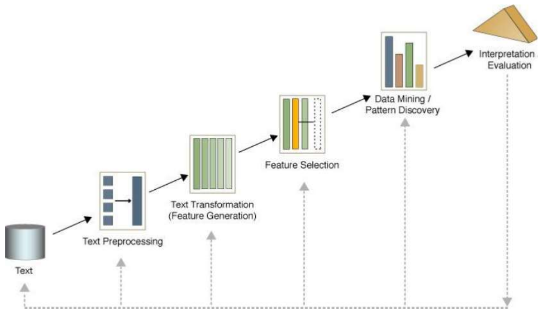


Introduction :Text mining Process

Preprocessing
 Lexical/Syntactic/Semantic analysis/Parsing

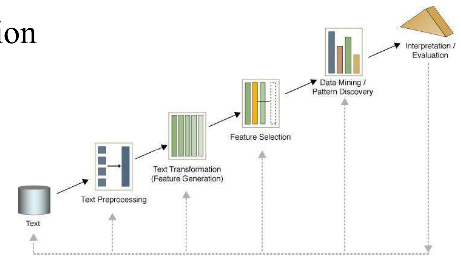
Features Extraction Indexing
 Stop words elimination
 Stemming-lemmatization

Weighting models
 Boolean
 Term frequency
 tf-iddf
 Entropy based model
 Similarity measure (cosinus)



Introduction :Text mining Process

Features Extraction - Dimension reduction
 Frequency
 Khi2
 LSA
Text/Data Mining
 Association rules
 Classification- Supervised learning - categorization
 Clustering- Unsupervised learning
 Link analysis
 Sequence analysis



Introduction :Text mining books

- Weiss S., Indurkhya N., Zhang T., Damerau F., « Text Mining Predictive methods for analyzing unstructured information », Springer, 2005.
- Feldman R., Sanger J., « The text mining handbook Advanced approaches in analyzing unstructuerd data », Cambridge University Press ,
- Aggarwal C., Zhai C., « Mining Text Data », Springer, 2012.
- Ignatow G., Mihalcea R., « Text mining A guidebook for the social science », Sage Publications, 2016.
- Perkins J., « Python text processing with NLTK 2.0 cookbook », Packt Publising ,
- Ted Kwartler, Text Mining in Practice with R, 2017
- B.Bengfort , R. Bilbro , T. Ojeda, Applied Text Analysis with Python: Enabling Language-Aware Data Products with Machine Learning, 2018
- A. Géron Hands-On Machine Learning with Scikit-Learn and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems, 2017

Outline

- Introduction
- Preprocessing
- Features extraction - indexing
- Weighting models
- Features Extraction - dimension reduction
- Text mining (association - categorization – clustering)
- References

- Application with R

Preprocessing

Task: Extract the words from the texts

- Remove unnecessary data (images, tables, …)
- Identify the encoding of the characters (Iso-Latin, UTF-8,…) to read the text

- Lexical analysis or tokenization
- Syntactic analysis or tagging
- Parsing

Preprocessing

Task: Remove unnecessary data (images, tables, …)

Text

Ohio Mattress Co said its first\nquarter, ending February 28, profits may be below the 2.4 mln\ndlrs, or 15 cts a share, earned in the first quarter of fiscal\n1986.\n The company said any decline would be due to expenses\nrelated to the acquisitions in the middle of the current\nquarter of seven licensees of Sealy Inc, as well as 82 pct of\nthe outstanding capital stock of Sealy.\n Because of these acquisitions, it said, first quarter sales\nwill be substantially higher than last year's 67.1 mln dlrs.\n Noting that it typically reports first quarter results in\nlate march, said the report is likely to be issued in early\nApril this year.\n It said the delay is due to administrative considerations,\nincluding conducting appraisals, in connection with the\nacquisitions.\n Reuter

lower casse, remove punctuation, numbers, \n, space non useful

Text after cleaning

ohio mattress co said its first quarter ending february profits may be below the mln dlrs or cts a share earned in the first quarter of fiscal the company said any decline would be due to expenses related to the acquisitions in the middle of the current quarter of seven licensees of sealy inc as well as pct of the outstanding capital stock of sealy because of these acquisitions it said first quarter sales will be substantially higher than last years mln dlrs noting that it typically reports first quarter results in late march said the report is likely to be issued in early april this year it said the delay is due to administrative considerations including conducting appraisals in connection with the acquisitions reuter

Preprocessing (Lexical)

- Lexical analysis (Tokenization) : recognition of the words
  - characters
  - words -> delimitors (space…) or use dictionary (Wordnet)
  - sets of words : named entity, ngrams, cooccurences, collocations (Pomme de terre, swimming pool)
  - tags : xml, html

Un accord Microsoft-Yahoo peut-être ce week-end
[2008-05-03 09:35]
NEW YORK (Reuters) - Microsoft et Yahoo négocient et leurs discussions se sont même intensifiées pour tenter de parvenir à un accord amiable par lequel le groupe internet passerait dans le giron du numéro un mondial des logiciels, a-t-on indiqué vendredi de sources proches du dossier.

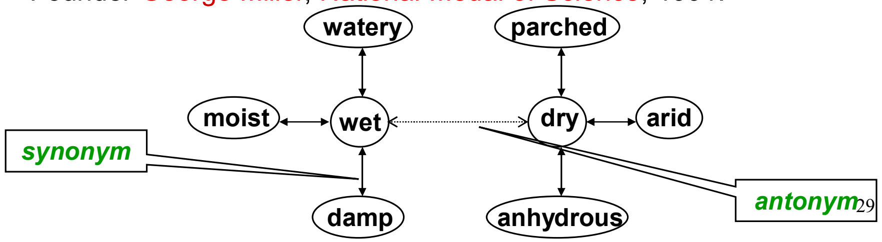
Preprocessing

- Lexical analysis with dictionary like WordNet

http://wordnet.princeton.edu/

An extensive lexical network for the English language

- Contains over 138,838 words.
- Several graphs of synsets, one for each part-of-speech.
- *Synsets* (synonym sets), each defining a semantic sense.
- Relationship information : super-subordinate relation (also called hyperonymy, hyponymy, or ISA relation, antonym)
- Downloadable for free (UNIX, Windows)
- Expanding to other languages (Global WordNet Association)
- Funded >$3 million, mainly government (translation interest)
- Founder George Miller, National Medal of Science, 1991.



Preprocessing (syntactic)

- Syntactic analysis/ part of speech (POS) tagging / tagger (étiquetage morpho-syntaxique): associate a grammatical category to each word

Main word category

- Noun
- Verb
- Adjective
- Adverb
- Pronoun
- Preposition
- Conjunction
- Determiner
- Exclamation

Brown/Penn Treebank tags

Tag	Description	Example	Tag	Description	Example
CC	Coordin. Conjunction	*and, but, or*	SYM	Symbol	+,%, &
CD	Cardinal number	*one, two, three*	TO	"to"	*to*
DT	Determiner	*a, the*	UH	Interjection	*ah, oops*
EX	Existential 'there'	*there*	VB	Verb, base form	*eat*
FW	Foreign word	*mea culpa*	VBD	Verb, past tense	*ate*
IN	Preposition/sub-conj	*of, in, by*	VBG	Verb, gerund	*eating*
JJ	Adjective	*yellow*	VBN	Verb, past participle	*eaten*
JJR	Adj., comparative	*bigger*	VBP	Verb, non-3sg pres	*eat*
JJS	Adj., superlative	*wildest*	VBZ	Verb, 3sg pres	*eats*
LS	List item marker	*1, 2, One*	WDT	Wh-determiner	*which, that*
MD	Modal	*can, should*	WP	Wh-pronoun	*what, who*
NN	Noun, sing. or mass	*llama*	WP$	Possessive wh-	*whose*
NNS	Noun, plural	*llamas*	WRB	Wh-adverb	*how, where*
NNP	Proper noun, singular	*IBM*	$	Dollar sign	$
NNPS	Proper noun, plural	*Carolinas*	#	Pound sign	#
PDT	Predeterminer	*all, both*	"	Left quote	(' or ")
POS	Possessive ending	*'s*	"	Right quote	(' or ")
PP	Personal pronoun	*I, you, he*	(	Left parenthesis	([, (, {, <)
PP$	Possessive pronoun	*your, one's*	)	Right parenthesis	( ],), }, >)
RB	Adverb	*quickly, never*	,	Comma	,
RBR	Adverb, comparative	*faster*	.	Sentence-final punc	(. ! ?)
RBS	Adverb, superlative	*fastest*	:	Mid-sentence punc	(: ; ... – -)
RP	Particle	*up, off*			

Preprocessing

- Syntactic analysis/ part of speech (POS) tagging / tagger (étiquetage morpho-syntaxique): associate a grammatical category to each word

- Exemple in french

Exemple

la séance est ouverte à 15h43.

phrase	étiquettes
la	Det+Feminin+Singulier
séance	NomCommun+Feminin+Singulier
est	Verbe+IndPré+Singulier+3Pers
ouverte	Verbe+ParPas+feminin+Singulier
à	Prep
15	Num
h	NC
43	Num
.	Ponc

Preprocessing

- Syntactic analysis/ part of speech (POS) tagging / tagger (étiquetage morpho-syntaxique): associate a grammatical category to each word

- Exemple in english

- *The grand jury commented on a number of other topics .*

⇨ The/DT grand/JJ jury/NN commented/VBD on/IN a/DT number/NN of/IN other/JJ topics/NNS ./.

Preprocessing

- Syntactic analysis with a dictionary

Exemple de Lexique avec étiquettes

```
avions.N
avions.V
ferme.N
ferme.V
ferme.A
belle.A
belle.N
```

Lean: adjective
Lean: verb

The lean man stopped to lean against the post (lean: adjective or verb)

Preprocessing

- Syntactic analysis is difficult :

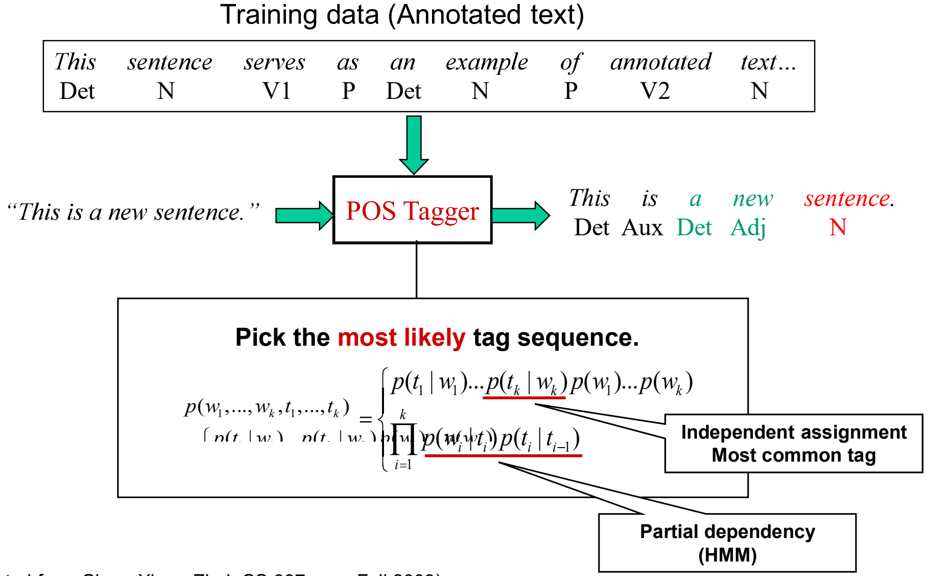
Sherlock saw the man with binoculars

Sherlock saw the man with binoculars

word/sentence sense desambiguisation

Preprocessing

- Syntaxic analysis with a tagger / training set



(Adapted from ChengXiang Zhai, CS 397cxz – Fall 2003)

Preprocessing

- Syntaxic analysis with Brill tagger
  - method for doing part-of-speech tagging
  - described by Eric Brill in his 1995 PhD thesis
  - error-driven transformation-based tagger: supervised learning

Ref: Eric Brill. 1992. A simple rule-based part of speech tagger. In Proceedings of the third conference on Applied natural language processing (ANLC '92)

Preprocessing

- Parsing: a dog is chasing a boy on the playground

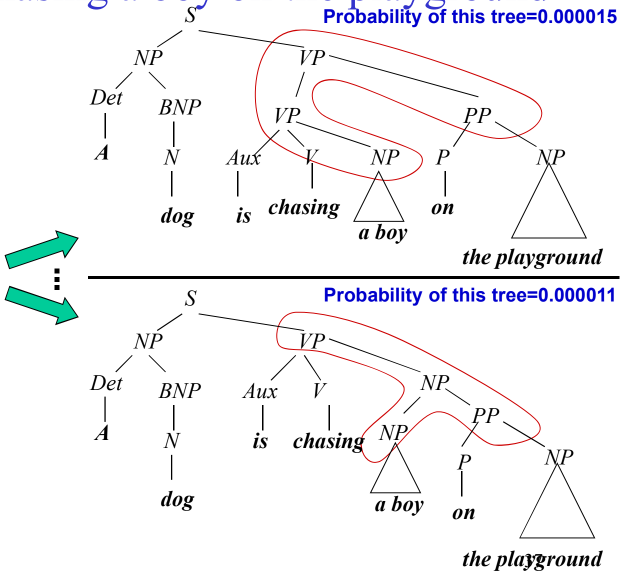
Probabilistic CFG

Grammar

S → NP VP	1.0
NP → Det BNP	0.3
NP → BNP	0.4
NP → NP PP	0.3
BNP → N	…
VP → V	
VP → Aux V NP	…
VP → VP PP	
PP → P NP	1.0

Lexicon

V → chasing	0.01
Aux → is	
N → dog	0.003
N → boy	
N → playground	…
Det → the	
Det → a	…
P → on	



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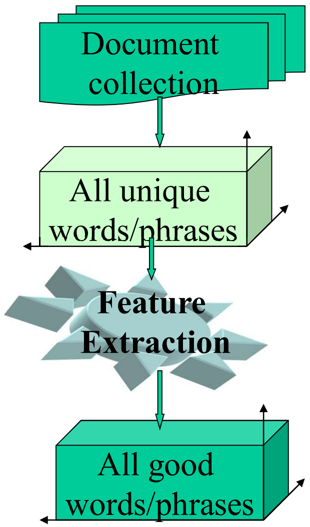
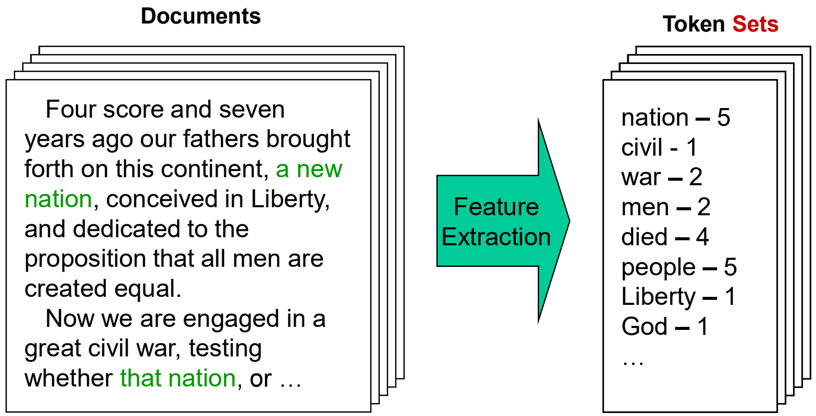
Features extraction - indexing

- Task : extract a good subset of words to represent documents
  - Stop words elimination to remove non informative words (the , and,..)
  - Stemming and lemmatization
  - Term weighting

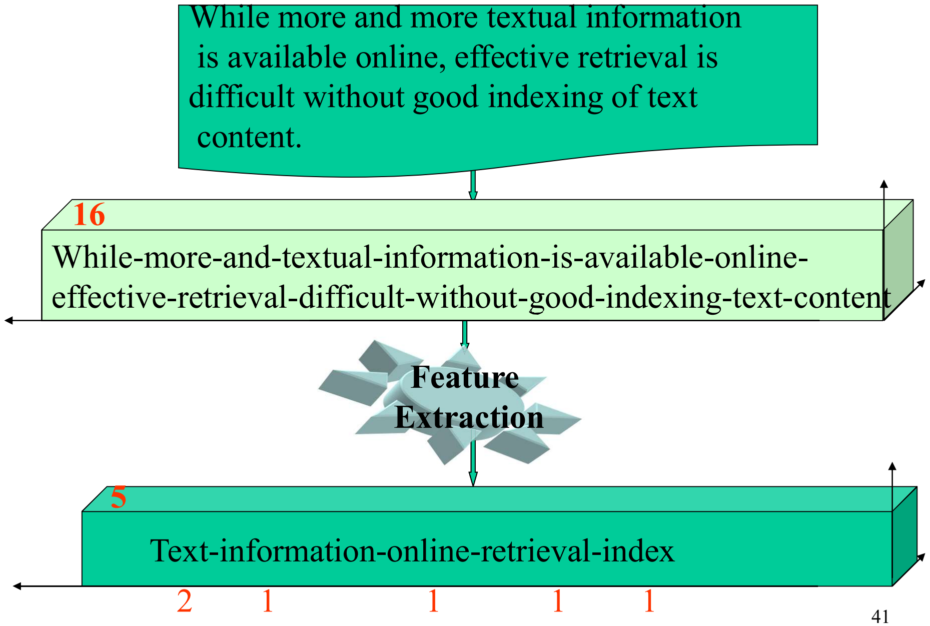


Features extraction - indexing



How can we do it ?

Features extraction - indexing



Features extraction - indexing

- Stop words elimination to remove non informative word (the , and,..) using
  - stop word list
  - statistical approach

Le Monde (février 2007), total de 1 286 915 occurrences :
de (72395), la (40558), le (31647), les (26237), et (24676), des (22621), en (20647), ..., est (13034), f´evrier (2851), deux (3473), ..., pays (1603), pr´esident (1484), ..., affairisme (1)
Le Petit Prince, 13 523 mots :
le (421), de (382), il (282), un (273), et (256), ..., petit (187), prince (163), ne (151), que (148), une (136), ... plan`ete (59), ..., auxquels (1).

Features extraction - indexing

- Stop words elimination with stop word list
  - French stop word list
    - http://snowball.tartarus.org/algorithms/french/stop.txt
  - English stop word list and in other languages
    - https://dev.mysql.com/doc/refman/5.1/en/fulltext-stopwords.html
    - http://www.ranks.nl/stopwords

Oracle text: 200 stop words in French

a , beaucoup, comment, encore, lequel, moyennant, près, ses, toujours, afin, ça, concernant, entre, les, ne, puis, sien, tous, ailleurs, ce, dans, et, lesquelles, ni, puisque, sienne, toute, ainsi, ceci, de, étaient, lesquels, non, quand, siennes, toutes, alors, cela, dedans, était, leur, nos, quant, siens, très, après, celle, dehors, étant, leurs, notamment, que, soi, trop, attendant, celles, déjà, etc, lors, notre, quel, soi-même, tu, au, celui, delà, eux, lorsque, notres, quelle, soit, un, aucun, cependant, depuis, furent, lui, nôtre, quelqu'un, sont, une, aucune, certain, des, grâce, ma, nôtres, quelqu'une, suis, vos, au-dessous, certaine, desquelles, hormis, mais, nous, quelque, sur, votre, au-dessus, certaines, desquels, hors, malgré, nulle, quelques-unes, ta, vôtre, auprès, certains, dessus, ici, me, nulles, quelques-uns, tandis, vôtres, auquel, ces, dès, il, même, on, quels, tant, vous, aussi, cet, donc, ils, mêmes, ou, qui, te, vu, aussitôt, cette, donné, jadis, mes, où, quiconque, telle, y, autant, ceux, dont, je, mien, par, quoi, telles, autour, chacun, du, jusqu, mienne, parce, quoique, tes, aux, chacune, duquel, jusque, miennes, parmi, sa, tienne, auxquelles, chaque, durant, la, miens, plus, sans, tiennes, auxquels, chez, elle, laquelle, moins, plusieurs, sauf, tiens, avec, combien, elles, là, moment, pour, se, toi, à, comme, en, le, mon, pourquoi, selon, ton.

List of stop words in English

a,able,about,across,after,all,almost,also,am,among,an,and,any,are,as,at,be,because,been,but,by,can,cannot,could,dear,did,do,does,either,else,ever,every,for,from,get,got,had,has,have,he,her,hers,him,his,how,however,i,if,in,into,is,it,its,just,least,let,like,likely,may,me,might,most,must,my,neither,no,nor,not,of,off,often,on,only,or,other,our,own,rather,said,say,says,she,should,since,so,some,than,that,the,their,them,then,there,these,they,this,tis,to,too,twas,us,wants,was,we,were,what,when,where,which,while,who,whom,why,will,with,would,yet,you,your

Features extraction - indexing

- Stop words elimination to remove non informative word (the , and,..) using statistical approach

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Le Petit Prince, 13 523 mots :
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Features extraction - indexing

- Stop words elimination

Text

ohio mattress co said its first quarter ending february profits may be below the mln dlrs or cts a share earned in the first quarter of fiscal the company said any decline would be due to expenses related to the acquisitions in the middle of the current quarter of seven licensees of sealy inc as well as pct of the outstanding capital stock of sealy because of these acquisitions it said first quarter sales will be substantially higher than last years mln dlrs noting that it typically reports first quarter results in late march said the report is likely to be issued in early april this year it said the delay is due to administrative considerations including conducting appraisals in connection with the acquisitions reuter

125 terms

Text after stp word elimination

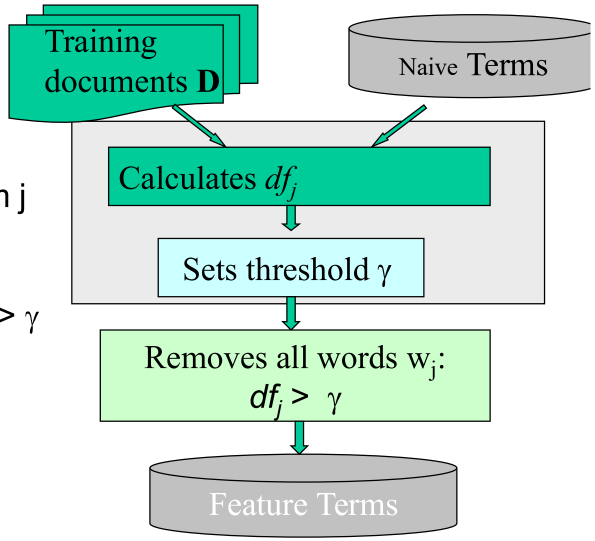
ohio mattress co said first quarter ending february profits may mln dlrs cts share earned first quarter fiscal company said decline due expenses related acquisitions middle current quarter seven licensees sealy inc well pct outstanding capital stock sealy acquisitions said first quarter sales will substantially higher last years mln dlrs noting typically reports first quarter results late march said report likely issued early april year said delay due administrative considerations including conducting appraisals connection acquisitions reuter

76 terms

$\frac{125-76}{125}$ = 39.2%terms removed

Dimension Reduction:DocFrequency Thresholding

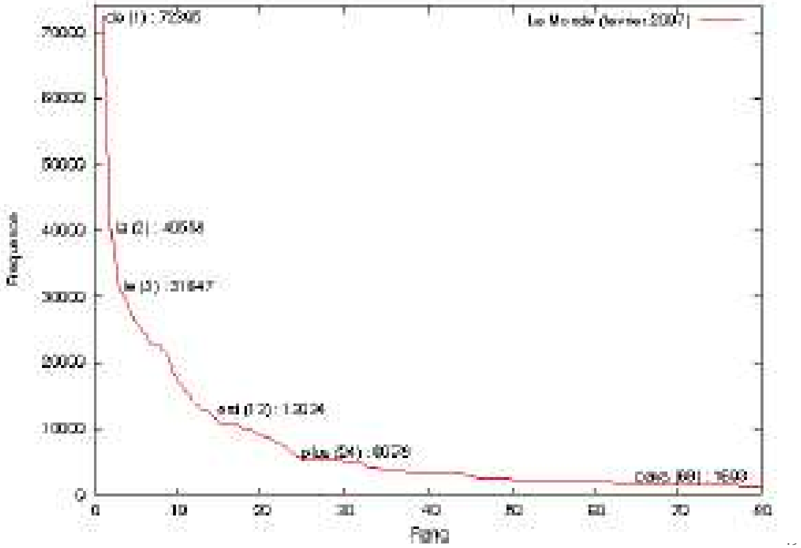
- Calculates document frequency dfj for each term j in training collection.
- Sets a threshold γ and removes all terms if its $df_j > \gamma$
- It is the simplest method with the lowest cost in computation.



Ref:[11][20][21][27]

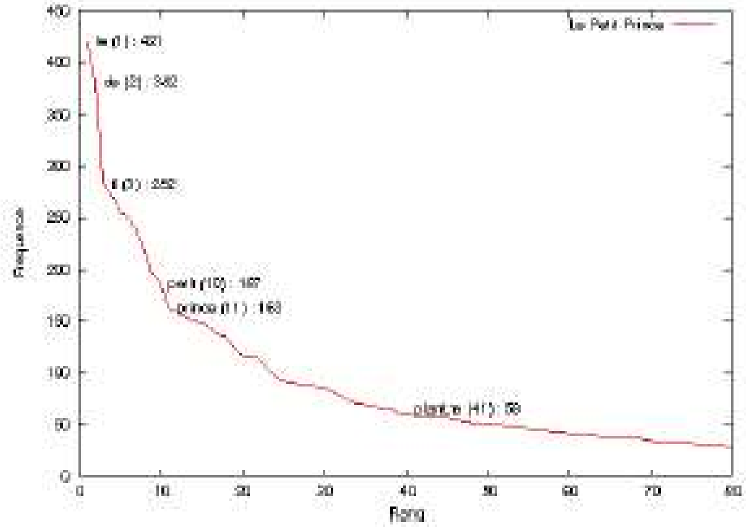
Features extraction - indexing

- Stop words elimination – Le monde



Features extraction - indexing

- Stop words elimination – Le petit prince



Features extraction - indexing

- Stop words elimination based on Zipf law
  - Empirical law introduced in 1936 at Haward by George Kingsley Zipf (1902-1950)
  - If the words are sorted by decreasing order of frequency, the frequency of a word is inversely proportional to its rank.

  ->

  frequency of the word at the rank k is approximately proportional to the reciprocal of k (1/k)

Features extraction - indexing

•Zipf law

Petit retour sur la loi de Zipf

$$f(r; s, C) = \frac{C}{r^s}, C \text{ constante}, s \approx 1$$

→ Given some corpus of natural language, the frequency f(r;s,C) of any word is inversely proportional to its rank r in the frequency table (where C is a constante and s ≈ 1).

Brown Corpus (> 1 M mots)

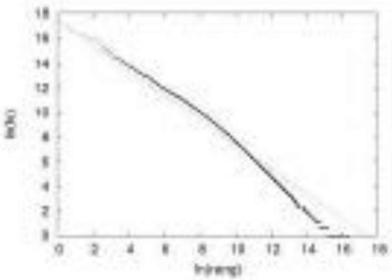
Word	Rank	Frequency	%
the	1	69971	7%
of	2	36411	3.5 %
and	3	28852	2.6%

Features extraction - indexing

•Zipf law with a logarithimic scale (ln)

Petit retour sur la loi de Zipf

$$f(r; s, C) = \frac{C}{r^s}, C \text{ constante}, s \approx 1$$



rang	mot	fréquence
1	*de*	36 875 868
2	*la*	16 565 726
3	*le*	12 639 034
4	*et*	11 587 487
5	*en*	10 885 221
6	*l'*	8 937 203
7	*du*	8 541 846
8	*des*	8 302 026
9	*d'*	7 766 576
10	*les*	7 722 951

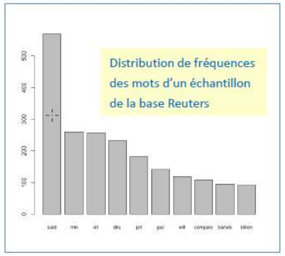
Fig. 2.3 – Illustration de la loi de Zipf sur la collection du Wikipédia français (gauche). Pour plus de visibilité nous avons utilisé une double échelle logarithmique, où ln est le logarithme népérien. La droite qui interpole le mieux les points, au sens des moindres carrés, est d'équation ln(fc) = 17.42 − ln(rang). A droite, nous avons reporté les mots de la collection dont le rang vaut au plus dix, ainsi que leurs fréquences.

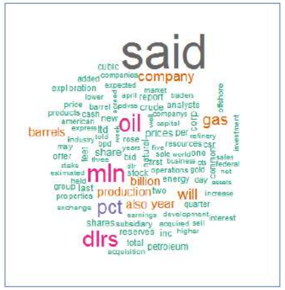
@E Gaussier

Features extraction - indexing

Zipf law and cloud of words

a good way to represent the content of a document





Features extraction - indexing

Stemming and lemmatization:

reduce inflectional forms and derivationally related forms of a word to a common base form

- stemming by heuristic process : cut the ends of words, removal of derivational affixes
- Without knowledge of the context

(F)	Rule			Example		
	SSES	→	SS	caresses	→	caress
	IES	→	I	ponies	→	poni
	SS	→	SS	caress	→	caress
	S	→		cats	→	cat

In a set of rules only one is obeyed, and this will be the one with the longest matching S1 for the given word.

Porter's algorithm (Porter, 1980)

http://www.tartarus.org/~martin/PorterStemmer/

Features extraction - indexing

Stemming and lemmatization:

- Lemmatization: properly use of a vocabulary and morphological analysis of words to return the dictionary form of a word known as the ***lemma***

Features extraction - indexing

Stemming and lemmatization example:

- walking → walk (Stem = lemma)
- better → good (lemma based on a dictionary)

Features extraction - indexing

Stemming and lemmatization:

ohio mattress co said first quarter ending february profits may mln dlrs cts share earned first quarter fiscal company said decline due expenses related acquisitions middle current quarter seven licensees sealy inc well pct outstanding capital stock sealy acquisitions said first quarter sales will substantially higher last years mln dlrs noting typically reports first quarter results late march said report likely issued early april year said delay due administrative considerations including conducting appraisals connection acquisitions reuter

before stemming : 549 caracters

ohio mattress co said first quarter end februari profit may mln dlrs cts share earn first quarter fiscal compani said declin due expens relat acquisit middl current quarter seven license seali inc well pct outstand capit stock seali acquisit said first quarter sale will substanti higher last year mln dlrs note typic report first quarter result late march said report like issu earli april year said delay due administr consider includ conduct apprais connect acquisit reuter

After stemming : 477 caracters

Features extraction - indexing

Stemming and lemmatization:

Document	imaging	databases	can	get	huge	most	save	images	permanently	store	baby
1	1	1	1	1	1	0	0	0	0	0	0
2	1	1	0	0	0	1	1	1	1	0	0
3	1	1	0	0	0	0	0	1	0	1	1
4	3	3	0	0	0	0	0	3	0	3	0

initial dictionary = {baby, can, databases, get, huge, images, imaging, most, permanently, save, store} ➔ 11 terms

Dictionnary after stop word removing = {baby, can, databases, get, huge, images, imaging, permanently, save, store} ➔ 10 terms

Dictionnary after stemming = {babi, can, databas, get, huge, imag, perman, save, store} ➔ 9 termes

Document	imag	databas	can	get	huge	save	perman	store	babi
1	1	1	1	1	1	0	0	0	0
2	2	1	0	0	0	1	1	0	0
3	2	1	0	0	0	0	0	1	1
4	6	3	0	0	0	0	0	1	0

Features extraction - Indexing and weighting models

- Index : bag of words used to represent the documents of the collection
- Weighting models: vector space models used to represent the documents

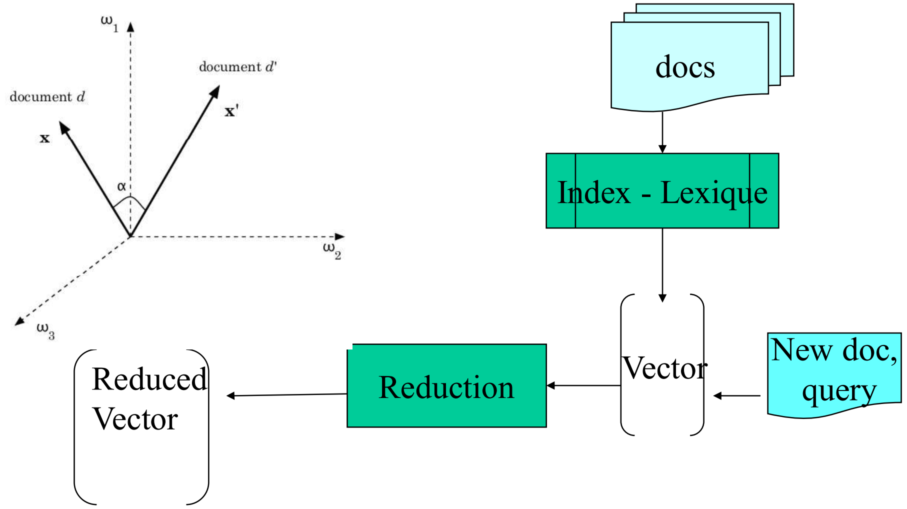
$$d_i=(w_{i1},w_{i2},\ldots w_{it})\in\mathbf{R}^t$$

w_{ij} is the weight of *j*th term in document d_i *where j belongs to the index Composed of t terms.*

Outline

- Introduction
- Preprocessing
- Features extraction - indexing
- Weighting models
- Features Extraction - dimension reduction
- Text mining (association - categorization – clustering)
- References
- Application with R

Features extraction - Indexing and weighting models



Weighting models : boolean model

- Based on the appearance of the index words in the document

$$d_i=(w_{i1},w_{i2},\dots w_{it})\in\mathbf{R}^t$$

w_{ij} is the weight of jth term in document d_i.

$$w_{ij} = \begin{cases} 1 & \text{if term j occurs in document } d_i, \\ 0 & \text{otherwise} \end{cases}$$

Weighting models : boolean model

- Based on the appearance of the index words in the document
- Example with an index T

T = {**chat, oiseau, ami, chien, ennemi**, **poisson, tortue**}

Le **chat** et **l'oiseau** sont **amis** mais le **chien** et l'**oiseau** sont **ennemis**

$\longrightarrow$ di = (1,1,1,1,1,0,0)

- Drawback: without reflection of importance of the word i.e. number of occurrences.

Weighting models : boolean model

Example

4 documents (Coelho & Richert, 2015 ; page 55,

1. Imaging databases can get huge.
2. Most imaging databases save images permanently.
3. Imaging databases store images baby.
4. Imaging databases store images. Imaging databases store images. Imaging databases store images.

Weighting models : boolean model

Example

4 documents (Coelho & Richert, 2015 ; page 55,

1. Imaging databases can get huge.
2. Most imaging databases save images permanently.
3. Imaging databases store images baby.
4. Imaging databases store images. Imaging databases store images. Imaging databases store images.

1. image databases can get huge
2. most image databases store image permanently
3. image databases store image baby
4. image databases store image image databases store image image databases store image

Document	databases	huge	image	permanently	store
1	1	1	1	0	0
2	1	0	1	1	1
3	1	0	1	0	1
4	1	0	1	0	1

Other drawbacks :

- the size of the index is not not known in advance
- the number of null values depends on the size of the document

Weighting models : TF model

•Based on the number of occurrences of the words (tf : Term Frequency weighting)

$$d_i=(w_{i1},w_{i2},\ldots w_{it})\in\mathbf{R}^t$$

$\mathbf{w_{ij}}$:the number of times jth term occurs in document d_i

$$\mathbf{w_{ij} = tf_{ij}}$$

Weighting models : TF model

•Based on the number of occurences of the words (tf : Term Frequency weighting)

$$d_i=(w_{i1},w_{i2},\ldots w_{it})\in\mathbf{R}^t$$

• Example with an index T

T = {**chat, oiseau, ami, chien, ennemi**, **poisson, tortue**}

Le **chat** et **l'oiseau** sont **amis** mais le **chien** et l'**oiseau** sont **ennemis** ⟶ di = (1,2,1,1,1,0,0)

•

Weighting models : boolean model / TF

Example

1. image databases can get huge
2. most image databases store image permanently
3. image databases store image baby
4. image databases store image image databases store image image databases store image

Document	databases	huge	image	permanently	store
1	1	1	1	0	0
2	1	0	2	1	1
3	1	0	2	0	1
4	3	0	6	0	3

Features extraction - TF model

tf : Term Frequency weighting

- $w_{ij} = tf_{ij}$

the number of times jth term occurs in document Di.

- **Drawback: without reflection of importance factor for document discrimination.**

•**Ex.**

D1: *ABRTSAQWA XAO*

D2: *RTABBAXA QSAK*

	A	B	K	O	Q	R	S	T	W	X
D1	4	1	0	1	1	1	1	1	1	1
D2	4	2	1	0	1	1	1	1	0	1

Ref:[11][22]

Features extraction - tf idf model

tf×idf - Inverse Document Frequency weighting [Salton 1989]

$$d_i=(w_{i1}, w_{i2}, \ldots w_{it}) \in \mathbf{R}^t$$

$$w_{ij} = tf_{ij} * idf_j$$

Inverse Document Frequency: $idf_j = \ln(N/df_j)$

N : the number of documents in the document collection.
df_j : the number of documents in which the jth term occurs
ln : natural logarithm

- Advantage: with reflection of importance factor for document discrimination.
- Assumption: Terms with low df are better discriminator than ones with high df in document collection
- Remark: several variants (see SMART Salton75)

Features extraction - tf idf model

Example

doc1: mining search Engine ….

doc2: travel text … map travel

doc3: government president Congress …

……

	text	mining	travel	map	data	engine	govern	president	congress
IDF(faked)	1.4	3	2.8	3.3	2.1	5.4	2.2	3.2	4.3
doc1	0	1(3)	0	0	0	1(5,4)	0	0	0
doc2	1(1.4)	0	2 (5.6)	1(3.3)	0	0	0	0	0
doc3	2(2.8)	0	0	0	0	0	1 (2.2)	1(3.2)	1(4.3)
newdoc	2(2.8)	0	0	0	1(2,1)	1(5,4)	0	0	0

According to tf, newdoc is more close to doc3
According to tf idf newdoc is more close to doc1

Features extraction - tf idf model

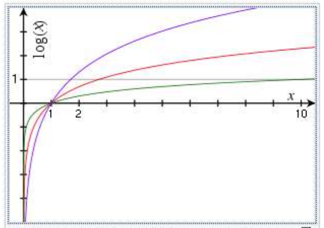
tf×idf - Inverse Document Frequency weighting

[Salton 1989]

Inverse Document Frequency: $idf_j = \log(N/df_j)$

N : the number of documents in the document collection.

df_j : the number of documents in which the jth term occurs



Fonctions logarithmes : en rouge la fonction logarithme de base e, en vert celle de base 10 et en violet celle de base 1,7.

Logarithm in IDF

- Gaussier : **ln natural logarithm**
- Sparck Jones : original measure where logarithm was specifically to the base 2
- Robertson (Understanding Inverse Document Frequency: On theoretical arguments for IDF,)

"the base of the logarithm is not in general important"

Weighting models: other variants

(Smart – Salton 2017)

- Variants of term frequency (tf) weight
  - Simple normalization : tf adjusted for document length
    $w_{ij} = tf_{ij} /$ (number of words in d_i)
  - Logarithmic normalization
    $w_{ij} =$ 0 si $tf_{ij} = 0$
    $1 + \log_{10}(tf_{ij})$ sinon
- Variants of inverse document frequency (idf) weight

  inverse document frequency smooth (avoid idf null)
  $idf_j = \ln(1 + N/df_j)$

Weighting models: variants for tf

Document	databases	huge	image	permanently	store
1	1	1	1	0	0
2	1	0	2	1	1
3	1	0	2	0	1
4	3	0	6	0	3

- Logarithmic normalization

Document	databases	huge	image	permanently	store
1	1.00	1.00	1.00	0.00	0.00
2	1.00	0.00	1.30	1.00	1.00
3	1.00	0.00	1.30	0.00	1.00
4	1.48	0.00	1.78	0.00	1.48

- Simple normalization : tf adjusted for document length

Document	databases	huge	image	permanently	store
1	0.33	0.33	0.33	0.00	0.00
2	0.20	0.00	0.40	0.20	0.20
3	0.25	0.00	0.50	0.00	0.25
4	0.25	0.00	0.50	0.00	0.25

Weighting models: Entropy weighting

$$w_{ij} = \log(tf_{ij} + 1.0) * (1 + entropy(w_i))$$

where

$$entropy(w_i) = \frac{1}{\log(N)} \sum_{j=1}^{N} \left[\frac{tf_{ij}}{df_j} \log\left(\frac{tf_{ij}}{df_j} \right) \right]$$

is average entropy of ith term and

-1: if word occurs once time in every document

0: if word occurs in only one document

➢ A term whose appearance is concentrated in few documents is given a higher weight in the documents where it appears

Ref:[13][22]

Weighting models: n grams

***n*-gram** : contiguous sequence of *n* items from a given text
Item can be word or character

Example of 4-grams extracted from « image databases can get huge»
-> ‘imag’, ‘mage’, ‘age‘, ‘ged’, ‘e da’, ‘ dat’, ‘data’, ‘atab’, …

Interest:

- langage independence
- like stemming
- good results for spam or plagiarism detection

Limit : complexity for n > 2

Weighting models : similarity measures

measure the closeness between two documents relative to term occurrences

- Given two documents

  $D_i = (w_{i1},\dots,w_{it})$ and $D_l = (w_{l1}, .., w_{lt})$

- Similarity measures
  - dot product

    $$Sim(D_i, D_l) = Sim(D_i, D_l) = \sum_{j=1,t} wij * wlj$$

  - normalized dot product (or cosine)

    $$Sim(D_i, D_l) = \frac{\sum_{j=1,t} wij * wlj}{\sqrt{\sum_{j=1,t}(wij)^2 * \sum_{j=1,t}(wlj)^2}}$$

Weighting models : similarity measures

Normalized dot product (or cosine)

$D_i = (w_{i1},\dots,w_{it})$ and $D_l = (w_{l1}, .., w_{lt})$

$$Sim(D_i, D_l) = \frac{\sum_{j=1,t} wij * wlj}{\sqrt{\sum_{j=1,t}(wij)^2 * \sum_{j=1,t}(wlj)^2}}$$

$Sim(D_i,D_j) \in [-1, 1]$
$Sim(D_i,D_j) \in [0, 1]$ with $w_{ij} > 0$

Weighting models : similarity measures

Normalized dot product (or cosine)

$D_i = (w_{i1},\dots,w_{it})$ and $D_l = (w_{l1}, .., w_{lt})$

$$Sim(D_i, D_l) = \frac{\sum_{j=1,t} wij * wlj}{\sqrt{\sum_{j=1,t}(wij)^2 * \sum_{j=1,t}(wlj)^2}}$$

Document	databases	huge	image	permanently	store
1	1	1	1	0	0
2	1	0	2	1	1
3	1	0	2	0	1
4	3	0	6	0	3tf

$$\cos(3,4) = \frac{1\times 3 + \cdots + 1\times 3}{\sqrt{1^2 + \cdots + 1^2} \times \sqrt{3^2 + \cdots 3^2}} = 1$$

Weighting models : similarity measures

Properties of similarity metric

Non-negativity $s(u,v) \geq 0 \;\; \forall u,v$

Symetric $s(u,v) = s(v,u) \;\; \forall u,v$

Maximality $s(u,u) = s(v,v) = s_{\max} \;\; \forall u \neq v$

Normalisation $\begin{cases} s(u,v) = 1 \Leftrightarrow u = v \;\; \forall u,v \\ s(u,v) < 1 \end{cases}$

Weighting models : similarity measures

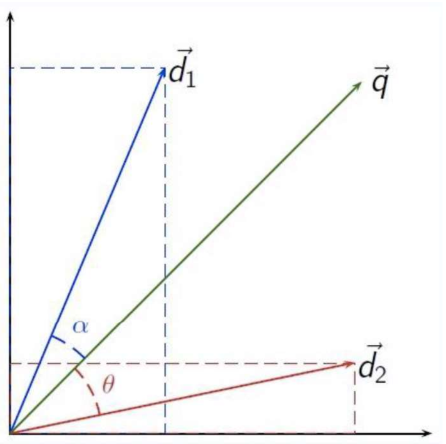
Dissimilarity from similarity metric

$\mathrm{Dist}(D_i,D_j) = 1-\mathrm{Sim}(D_i,D_j)$

$\mathrm{Dist}(D_i,D_j) = \begin{cases} 1-\mathrm{Sim}(D_i,D_j) \in [0, 1] \text{ if } w_{ij} > 0 \\ \mathrm{Smax} - \mathrm{Sim}(D_i,D_j) \text{ otherwise} \end{cases}$

Weighting models : similarity measures

Cosine measure



•$\alpha < \theta$

•$\cos(\alpha) > \cos(\theta)$

•**d_1** is more close to **q** than **d_2**

sim(d1,q) > sim(d2,q)

•Cosine-based similarity model can reflect the *relations between features*.

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Feature Extraction:Dimension Reduction

- Reduce the size of the index to avoid the curse of dimensionality (R Bellman,1961)
- When the index size increases, the volume of the representation space increases so fast that the dataset becomes sparse.
- Limit the use of algorithms needing an amount of time or memory that is exponential in the number of dimensions
- Solutions:
  - Change the algorithm
  - Pre-processing the data into a lower-dimensional form

Feature Extraction: Index Dimension Reduction

Illustration

4 documents (Coelho & Richert, 2015 ; page 55,

1. Imaging databases can get huge.
2. Most imaging databases save images permanently.
3. Imaging databases store images baby.
4. Imaging databases store images. Imaging databases store images. Imaging databases store images.

Termes

Documents

Document	imaging	databases	can	get	huge	most	save	images	permanently	store	baby
1	1	1	1	1	1	0	0	0	0	0	0
2	1	1	0	0	0	1	1	1	1	0	0
3	1	1	0	0	0	0	0	1	0	1	1
4	3	3	0	0	0	0	0	3	0	3	0

1. Imaging databases can get huge.
2. Most imaging databases save images permanently.
3. Imaging databases store images baby.
4. Imaging databases store images. Imaging databases store images. Imaging databases store images.

1. image databases can get huge
2. most image databases store image permanently
3. image databases store image baby
4. image databases store image image databases store image image databases store image

Document	databases	huge	image	permanently	store
1	1	1	1	0	0
2	1	0	2	1	1
3	1	0	2	0	1
4	3	0	6	0	3

Feature Extraction: Index Dimension Reduction

- Reduce the size of the index
  - Selection of the most significative words
  - Association of words into concept
  - Classical dimension reduction

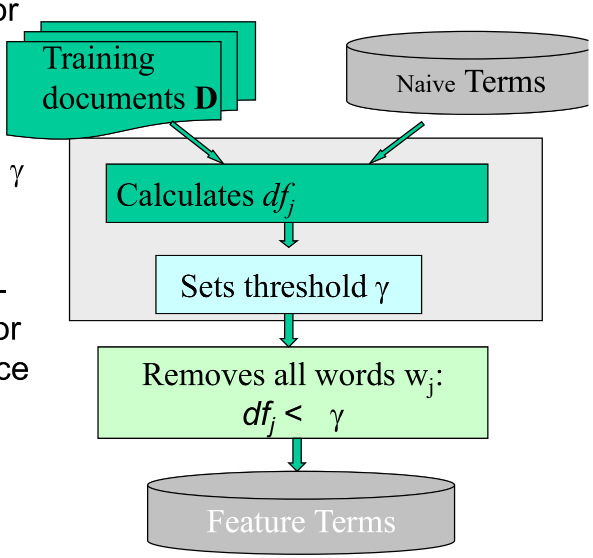
Feature Extraction:Dimension Reduction

- Document Frequency Thresholding
- X2-statistic
- Latent Semantic Indexing

Ref:[11][20][21][27]

Dimension Reduction: DocFreq Thresholding

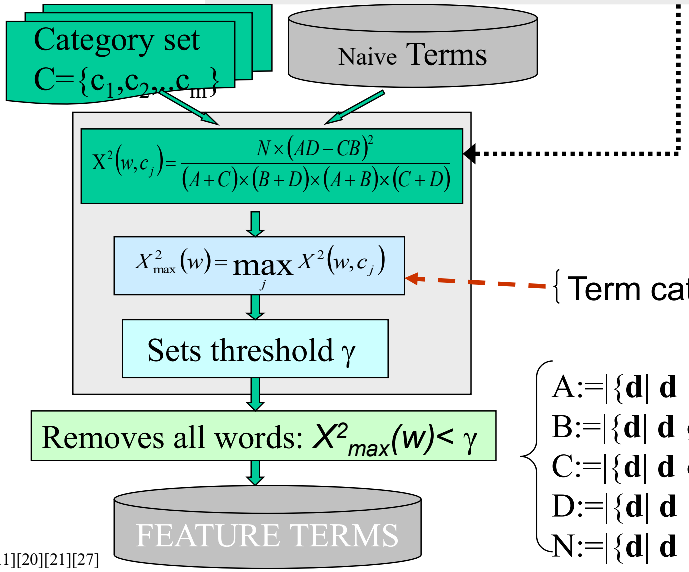
- Calculates frequency dfj for each term j in training collection.
- Sets a threshold γ and removes all terms if its dfj < γ holds.
(<> stop words dfj > γ)
- Rare terms are either non-informative for predictions or not influential in performance (<> in RI)
- It is the simplest method with the lowest cost in computation.



Ref:[11][20][21][27] dfj := the number of documents in which the jth term wj occurs

Dimension reduction: X2-statistic for classification task

- **Assumption**:a pre-defined category set for a training collection **D**
- **Goal**: Estimation independence between term and category



{Term categorical score}

$$\begin{cases} A := |\{\mathbf{d} \mid \mathbf{d} \in \mathbf{c_j} \wedge w \in \mathbf{d}\}| \\ B := |\{\mathbf{d} \mid \mathbf{d} \notin \mathbf{c_j} \wedge w \in \mathbf{d}\}| \\ C := |\{\mathbf{d} \mid \mathbf{d} \in \mathbf{c_j} \wedge w \notin \mathbf{d}\}| \\ D := |\{\mathbf{d} \mid \mathbf{d} \notin \mathbf{c_j} \wedge w \notin \mathbf{d}\}| \\ N := |\{\mathbf{d} \mid \mathbf{d} \in \mathbf{D}\}| \end{cases}$$

Ref:[11][20][21][27]

Dimension reduction: X2-statistic for classification task

- **Assumption**:a pre-defined category set for a training collection **D**
- **Goal**: Estimation independence between term and category

→ Use threshold γ

$$X^2(w,c_j)=\frac{N\times(AD-CB)^2}{(A+C)\times(B+D)\times(A+B)\times(C+D)}$$

$$\begin{cases} A := |\{\mathbf{d} \mid \mathbf{d} \in \mathbf{c_j} \wedge w \in \mathbf{d}\}| \\ B := |\{\mathbf{d} \mid \mathbf{d} \notin \mathbf{c_j} \wedge w \in \mathbf{d}\}| \\ C := |\{\mathbf{d} \mid \mathbf{d} \in \mathbf{c_j} \wedge w \notin \mathbf{d}\}| \\ D := |\{\mathbf{d} \mid \mathbf{d} \notin \mathbf{c_j} \wedge w \notin \mathbf{d}\}| \\ N := |\{\mathbf{d} \mid \mathbf{d} \in \mathbf{D}\}| \end{cases}$$

Removes all words: $X^2_{max}(w) < \gamma$

FEATURE TERMS

or

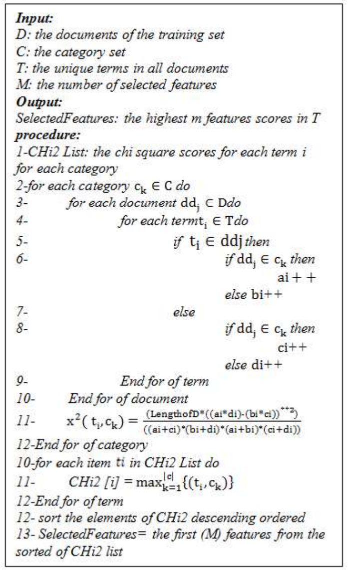
khi2 independence test (case of table 2x2)

Comparison with the theoretical value of X^2 (with 1 ddl)
If $X^2(w,c_j) > X^2$ is we can reject the hypothesis of independence
For instance if $X^2(w,c_j) > 3{,}84$ the test is *statistically significant* at the 0.05 level

Ref:[11][20][21][27]

Dimension reduction: X2-statistic for classification task

Mowafy, M., Rezk, A., El-Bakry, H.: An efficient classication model for unstructured text document. American Journal of Computer Science and InformationTechnology 06 (01 2018)



```
Input:
D: the documents of the training set
C: the category set
T: the unique terms in all documents
M: the number of selected features
Output:
SelectedFeatures: the highest m features scores in T
procedure:
1-CHi2 List: the chi square scores for each term i
for each category
2-for each category ck ∈ C do
3-      for each document ddj ∈ Ddo
4-            for each termti ∈ Tdo
5-                  if ti ∈ ddj then
6-                        if ddj ∈ ck then
                                 ai + +
                        else bi++
7-                  else
8-                        if ddj ∈ ck then
                                 ci++
                        else di++
9-            End for of term
10-     End for of document
11-     x2( ti,ck) = (LengthofD*((ai*di)-(bi*ci))**2) / ((ai+ci)*(bi+di)*(ai+bi)*(ci+di))
12-End for of category
10-for each item ti in CHi2 List do
11-     CHi2 [i] = max_{k=1}^{|c|}{(ti,ck)}
12-End for of term
12- sort the elements of CHi2 descending ordered
13- SelectedFeatures= the first (M) features from the
sorted of CHi2 list
```

Ref:[11][20][21][27]

Dimension Reduction:LSI

- LSA = Latent semantic Analysis also called Latent Semantic Indexing) (Deerwester, 1988)
  - based on SVD: Singular Values Decomposition of the rectangular matrix X(t,N) or X'(N,t)
  - related to PCA Principal Component Analysis (cf factorial analysis Cf M1.)

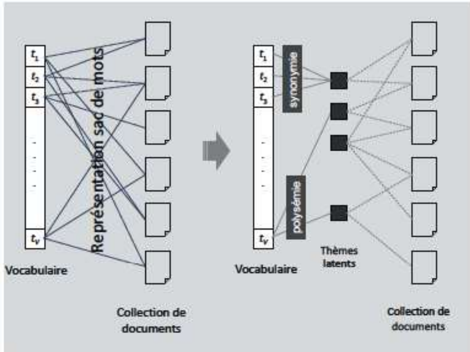
Synonymy: 2 words with the same meaning

big large

Polysemy: 1 word with different Meaning

Crane/grue: a bird or a construction equipment

-> latent factors



Dimension Reduction:LSI(1)

1.SVD Model : Singular Value Decomposition of matrix X(t,N)

$$\underset{\text{terms}}{\overset{\text{documents}}{\begin{pmatrix} X \end{pmatrix}}}_{(t,N)} = \begin{pmatrix} U \end{pmatrix}_{(t,t)} \bullet \begin{pmatrix} S^{*}\ {}^{*}{}_{*}{}_{*} \end{pmatrix}_{(t,N)} \bullet \begin{pmatrix} V' \end{pmatrix}_{(N,N)}$$

U: t × t orthogonal eigenvector matrix containing eigenvectors of *X X'*
V': transpose of V

V: NxN orthogonal eigenvector matrix containing eigenvectors of *X ' X*

S: t ×N diagonal matrix of singular values (square roots of eigenvalue of XX') in decreasing order of importance

$$\left[\begin{bmatrix} \mathbf{u}_1 \end{bmatrix} \cdots \begin{bmatrix} \mathbf{u}_l \end{bmatrix}\right] \cdot \begin{bmatrix} \sigma_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \sigma_l \end{bmatrix} \cdot \begin{bmatrix} [& \mathbf{v}_1 &] \\ & \vdots & \\ [& \mathbf{v}_l &] \end{bmatrix}$$

Ref:[11][20][21][27]

Dimension Reduction:LSA(2) : approximate model

Rank k approximation of X obtained by taking k largest singular values

$$\mathbf{X} \approx \begin{pmatrix} \mathbf{X}\hat{} \end{pmatrix}_{(t,N)} = \begin{pmatrix} \mathbf{Uk} \end{pmatrix}_{(t,k)} \bullet \begin{pmatrix} \mathbf{Sk} \end{pmatrix}_{(k,k)} \bullet \begin{pmatrix} \mathbf{Vk'} \end{pmatrix}_{(k,N)}$$

Select k<=min(t,N) !

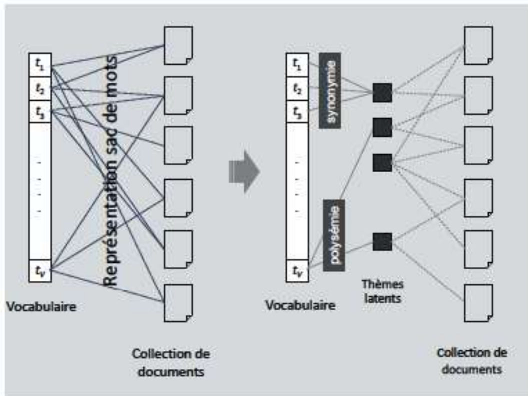
- S_k: diagonal matrix limited to the k largest singular values.
- U_k: k first colums of U
  - Every row of Uk approximatively represents a term on k latent factors
- V_k: k first colums of V
  - Every row of Vk represents a document on k latent factors
- Latent space
  - documents and terms are represented in the same space (with dimension k)

Ref:[11][20][21][27]

Dimension Reduction:LSA(2) : approximate model

- Uk: k first colums of U
  - Every row of Uk approximatively represents a term on k latent factors
- Vk: k first colums of V
  - Every row of Vk represents a document on k latent factors

!



Ref:[11][20][21][27]

Dimension Reduction:LSA(2) : approximate model

X^ : Rank k approximation of X with the smallest error (Frobenius)

$$||X - X^{\wedge}||_F = \sqrt{\sum_{i,j}(Xij \ - X^{\wedge}{}_{ij})^2}$$

Ref:[11][20][21][27]

Dimension Reduction:LSI

- Remarks :
  - Singular value decomposition can be done using Lanczos algorithm
  - Available in R
- Limits :
  - No good method to determine k. it depends on application domain
  - Experiments suggest to begin with law value (k=10) and to increase until a maximal value before to select the best value k*
- Extensions :
  - PLSA Probabilistic latent semantic analysis (Hofman et Puzicha1998)
  - Latent Dirichlet Allocation (LDA) - Topic models (Blei et al. 2003)
  - Word embedding (Word2vec and skip-gram – Mikolov2013)

Word embedding (Word2vec and skip-gram (Mikolov2013)

Aim: words or phrases from a corpus are mapped to vectors of real numbers.
How: using neural networks, dimensionality reduction on the word co-occurrence matrix, probabilistic models

Idea: words having the same context should be represented by similar vectors

Interest: Allow operation on the vectors such as
king – male + female = queen

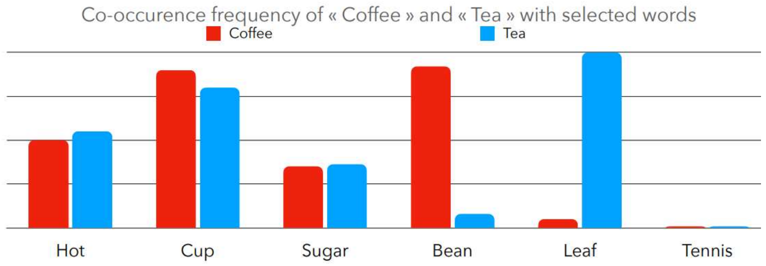
Limit: require large collection to learn the vectors associated to the words

Word embedding

Aim: words or phrases from a corpus are mapped to vectors of real numbers.
How: using neural networks, dimensionality reduction on the word co-occurrence matrix, probabilistic models

Hypothesis: Distributional structure (Harris, 1954)

- Words with similar meanings tend to occur in similar contexts



Word embedding

Aim: words or phrases from a corpus are mapped to vectors of real numbers.
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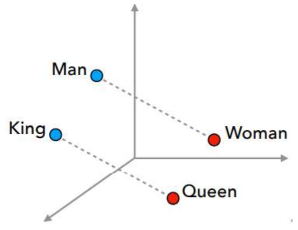
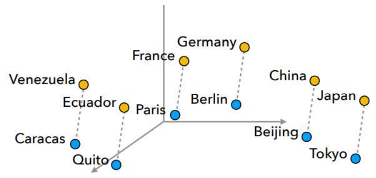


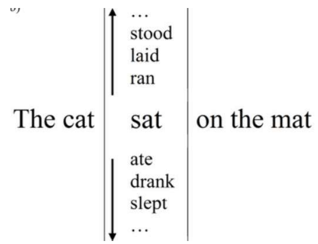


- Allows operation on the vectors such as
  King – Man + Woman = Queen

Word2Vec embedding model (Mokolov NIPS2013)

2 methods: CBOW (continuous bag of words) and **Skip-gram**

CBOW: to predict a word given its context
Skip-gram: to predict a context given a word



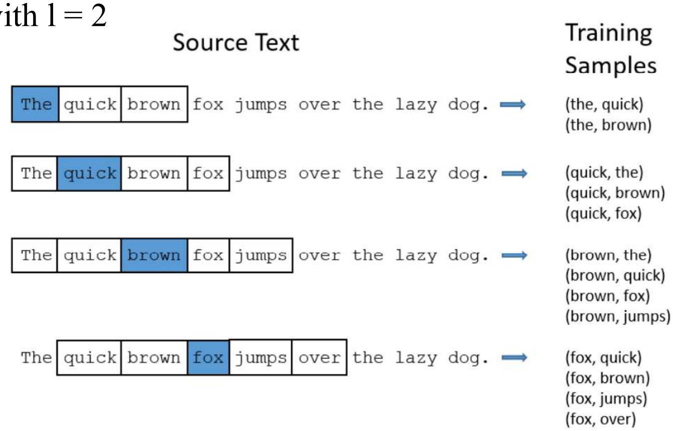
CBOW
The words cat and mat are given
CBOW should predict sat

Skip-gram
The word sat is given
Skip-gram should predict cat and mat

Word2Vec embedding model (Mokolov NIPS2013)

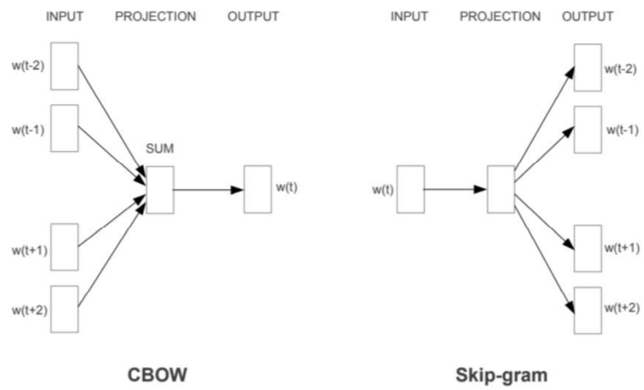
Given a corpus containing a vocabulary of n words
a sliding window of size l
1- Build a training set

For Skip-gram, with l = 2



Word2Vec embedding model (Mokolov NIPS2013)

2- Learn a vectorial representation of the words either as context or as target by solving an optimization problem with a neural network having one hidden layer



Word2Vec embedding model (Mokolov NIPS2013)

Skip-gram:
works well with a small amount of the training data
represents well even rare words or phrases.

CBOW:
several times faster to train than the skip-gram
slightly better accuracy for the frequent words.

Other embedding models

Glove *(J. Pennington, EMNLP, 2014)*
- exploits the sparce co-occurrence matrix which tabulates how frequentl
words co-occur in the corpus

FastText *(P. Bojanowski, TACL2017, Joulin ACL2017)*
- inspired by Word2vec
- but handle n-gram of characters instead of words
- allows to work with words out of the training corpus

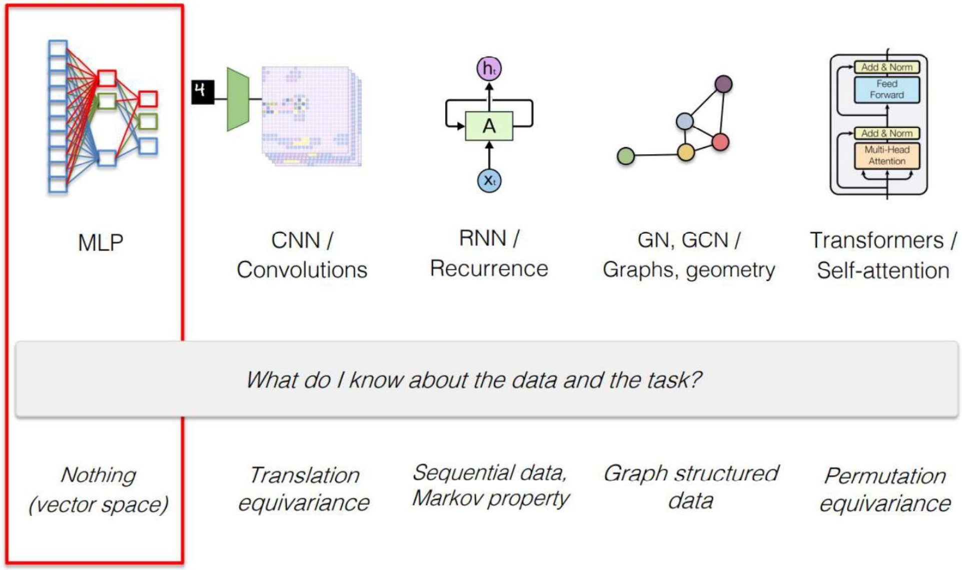
Other embedding models

ELMO: *Embedding from Language Models*
(Peters, NAACL2018)
- based on recurrent neural networks (LSTM - Long short-term memory

BERT: *Bidirectional Encoder Representations from Transformers*
(Delvin 2018)
- Allows to predict hidden words, next sentence, etc.
- From a pre-traiened embeddings learnt on a large generic corpus, possibility to rafine the embeddings on a smaller corpus for solving specific tasks.
- CamemBert, Flaubert: Versions dedicated to the French language

TRANSFORMER (*Bidirectional Encoder Representations from Transformers)*
(Vaswani Arxiv2017)

Various deep neural networks models



@ Wolf- Sleight2021

Outline

- Introduction
- Preprocessing
- Features extraction - indexing
- Weighting models
- Features Extraction - dimension reduction
- Text mining (association - categorization – clustering)
- References

- Application with R

Text mining tasks

- Keyword-based association analysis
- Automatic document classification
- Clustering - Similarity detection
- Link analysis: unusual correlation between entities
- Sequence analysis: predicting a recurring event
- Event/Novelty/Anomaly detection: find information that is new/violates usual patterns
- Author identification
- Hypertext analysis
  - Patterns in anchors/links
    - Anchor text correlations with linked objects

Text mining:keyword-based association analysis

- Motivation
  - Collect sets of keywords or terms that occur frequently together and then find the association or correlation relationships among them

- Association Analysis Process
  - Preprocess the text data by parsing, stemming, removing stop words, etc.
  - Evoke association mining algorithms (Apriori)
    - Consider each document as a transaction (<> vectorial representation)
    - View a set of keywords in the document as a set of items in the transaction
    - Detect association

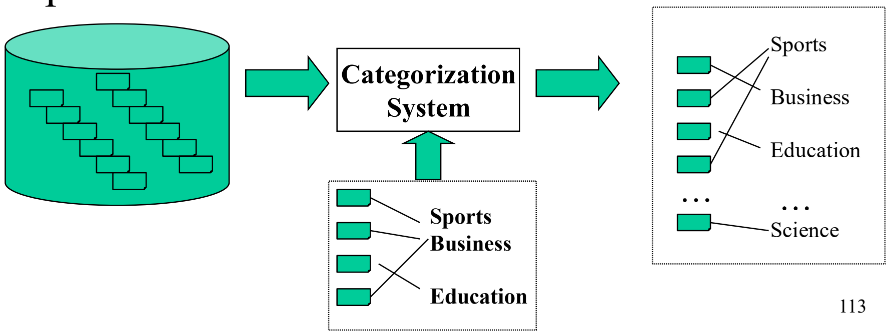
    (Example: Total => Gaz, Hubert Curien Laboratory => data mining)
  - Term level association mining
    - No need for human effort in tagging documents
    - The number of meaningless results and the execution time is greatly reduced

Text mining:text classification/categorization (supervised task)

- Motivation
  - Automatic classification for the large number of on-line text documents (Web pages, e-mails, corporate intranets, etc.)
- Classification Process
  - Data preprocessing
  - Definition of training set and test set
  - Creation of the classification model using the selected classification algorithm (decision tree, SVM, etc) and the training set
  - Classification model validation on the test set
  - Classification of new/unknown text documents

Text mining: text classification/categorizatio

- Pre-given categories and labeled document examples (Categories may form hierarchy)
- Classify new documents
- A standard classification (supervised learning) problem



Text mining: text classification

- Manual: Typically rule-based
  - Does not scale up (labor-intensive, rule inconsistency)
  - May be appropriate for special data on a particular domain
- Applications :
  - News article classification per topic (sports, culture, etc)
  - Automatic email filtering
    - Looking to send each message to the services concerned: complaint , billing , legal Service
  - Webpage classification
    - Assign a new page in Wikipedia categories

Text mining: text classification

Automatic: Typically exploiting machine learning techniques

- **Vector space model based**
  - Prototype-based (Rocchio)
  - K-nearest neighbor (KNN)
  - Decision-tree (learn rules)
  - Neural Networks (learn non-linear classifier)
  - Support Vector Machines (SVM)
- **Probabilistic or generative model based**
  - Naïve Bayes classifier
- **Many, many others and variants exist [F.S. 02]**
  - e.g. Bim, Nb, Ind, Swap-1, LLSF, Widrow-Hoff, Gis-W, …

Text mining: text classification

Model:Centroid-Based Classifier(1)

1.Input:new document $d=(w_1, w_2,\ldots,w_n)$;

2.Predefined categories:$C=\{c_1,c_2,\ldots,c_l\}$;

3.//Compute centroid vector

$$\vec{c}_i = \frac{\sum_{d' \in c_i} d'}{|c_i|} \quad ,\forall\ c_i \in C$$

4.//Similarity model - cosine function

$$Simil(d_i,d_j)=\cos(d_i,d_j)=\frac{d_i \bullet d_j}{\|d_i\|_2 \times \|d_j\|_2}=\frac{\sum w_{il} \times w_{jl}}{\sqrt{\sum w^2_{il}} \times \sqrt{\sum w^2_{jl}}}$$

5.//Compute similarity

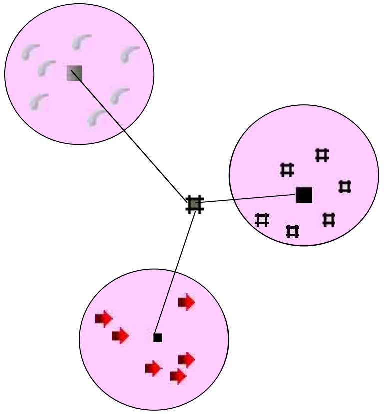
$$Simil(\vec{c}_i,d)=\cos(\vec{c}_i,d),\forall c_i$$

6.//Output:Assign to document d the **category** c_{max} such that:

$$Simil(\vec{c}_i,d) \le Simil(c_{\max},d)$$

Text mining: text classification

Model:Centroid-Based Classifier(1)



Text mining: text classification

Variant: Rocchio algorithm [37]

1.**Input****:new document** *d* $=(w_1, w_2, \ldots, w_l \ldots, w_n)$;

2.Predefined categories:$C=\{c_1, c_2, \ldots, c_k, \ldots, c_r\}$

3.Compute centroid vector for category c_k

$$c_{kl} = \alpha \frac{1}{|Ck|} \sum_{d_i \in ck} w_{il} - \beta \frac{1}{|D - C_k|} \sum_{d_i \notin Ck} w_{il} \qquad \forall\, c_k \in C \quad (1)$$

4.Similarity model - cosine function

$$Simil(d_i, d_j) = \cos(d_i, d_j) = \frac{d_i \bullet d_j}{\|d_i\|_2 \times \|d_j\|_2} = \frac{\sum w_{il} \times w_{jl}}{\sqrt{\sum w^2_{il}} \times \sqrt{\sum w^2_{jl}}}$$

5.Compute similarity

$$Simil(c_k, d) = \cos(c_k, d), \forall c_i$$

6.Output:Assign to document *d* the **category** c_{max}

$$Simil(c_k, d) \le Simil(c_{\max}, d)$$

7.Update the centroid by adding d to C_{max} and substracting d to the other classes according to formula (1)

Text mining: text classification

Model:K-Nearest Neighbor Classifier

1.Input:new document *d*;

2.training collection:$D=\{d_1, d_2, \ldots d_n\}$;

3.predefined categories:$C=\{c_1, c_2, \ldots, c_l\}$;

4.//Compute similarities

for($d_i \in D$){ **Simil**(d,d_i) =cos(d,d_i); }

5.//Select k-nearest neighbor

Construct k-document subset D_k such that

Simil(d,d_i) < min(**Simil**(d,doc) | doc $\in D_k$) $\forall d_i \in D- D_k$.

6.//Compute score for each category

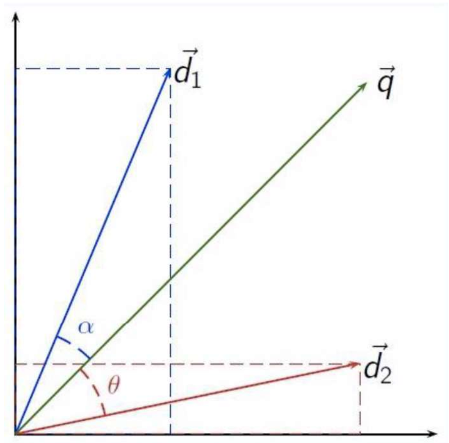
```
for(ci∈C){
    score(ci)=0;
    for(doc∈Dk){ score(ci)+=((doc∈ci)=true?1:0)}
}
```

7.//Output:Assign to *d* **the category** ***c* with the highest score:**

score(*c*) ≥ score(c_i) , $\forall c_i \in C- \{c\}$

3.3.2 Model:K-Nearest Neighbor Classifier

Cosine measure



- $\alpha < \theta$
- $\cos(\alpha) > \cos(\theta)$
- $\mathbf{d_1}$ is more close to **q** than $\mathbf{d_2}$

sim(d1,q) > sim(d2,q)

Model:Naive Bayes Classifier

Basic assumption: all terms are distributed in documents independently.

1.**Input**:**new document** *d;*

2.predefined categories:C={$c_1, c_2, \ldots, c_l$};

3.//Compute the probability that *d* **is in each class** *c* ∈C with the law of total probability

$$\Pr(c_i|d) = \frac{\Pr(d|c_i)\Pr(c_i)}{\Pr(d)} = \frac{\Pr(d|c_i)\Pr(c_i)}{\sum_{ck\in C}\Pr(d|c_k)\Pr(c_k)}$$

//note that terms w_i in document are independent each other

$$= \frac{\Pr(c_i)\prod_{j=1}^{|d|}\Pr(w_j|c_i)}{\sum_{c_k\in C}\left(\prod_{j=1}^{|d|}\Pr(w_j|c_k)\right)\Pr(c_k)}$$

4.//output:Assigns to *d* the **category** *c* with the highest probability:

$$\Pr(c|d) = \max_{ci\in C}(\Pr(c_i|d))$$

Text classification: evaluation

Accuracy, recall and precision on contingency table

	Belonging to C1(positive)	**Belonging** C2 (negative)	
Affected to C1 positive	a True positive Tp	b False positive Fp	a+b
Affected to C2 negative	c False negative Fn	d True negative Tn	c+d
	a+c	b+d	a+b+c+d

– Accuracy : a+d/(a+b+c+d)

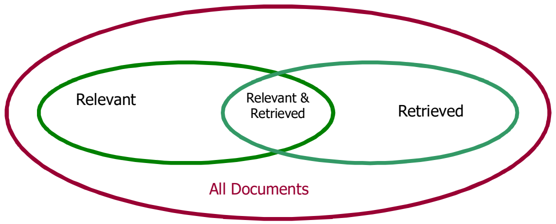
– Recall : r = a/(a+c)

– Précision : p = a/(a+b)

– Noise = 1-precision

– F-mesure , compromis between r and p: F1 = 2rp/(r+p)

Precision / recall (IR)



- Precision: the percentage of retrieved documents that are in fact relevant (i.e., "correct" responses)

$$precision = \frac{|\{Relevant\} \cap \{Retrieved\}|}{|\{Retrieved\}|}$$

- Recall: the percentage of documents that are relevant and were, in fact, retrieved

$$\text{recall} = \frac{|\{Relevant\} \cap \{Retrieved\}|}{|\{Relevant\}|}$$

Text mining: text classification

- Benchmarks
  – Classic: Reuters collection
    - A set of newswire stories classified under categories related to economics.
- Effectiveness
  – Difficulties of strict comparison
    - different parameter setting
    - different "split" (or selection) between training and testing
    - various optimizations … …
  – However widely recognizable
    - Best: Boosting-based committee classifier & SVM
    - Worst: Naïve Bayes classifier
  – Need to consider other factors, especially efficiency

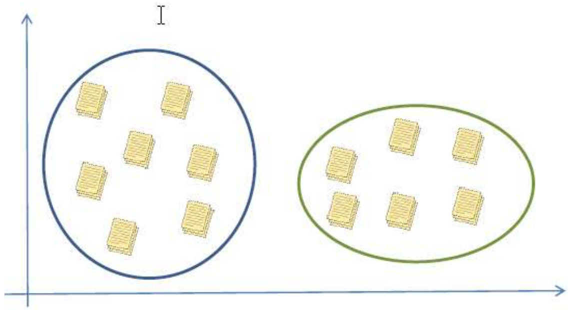
Text mining: text classification

- Summary text classification / categorization
  - Wide application domains
  - Comparable effectiveness to professionals
    - Manual TC is not 100% and unlikely to improve substantially.
    - A.T.C. is growing at a steady pace
  - Prospects and extensions
    - Very noisy text, such as text from Optical Character Recognition (OCR)
    - Speech transcripts

			#1	#2	#3	#4	#5
		# of documents	21,450	14,347	13,272	12,902	12,902
		# of training documents	14,704	10,667	9,610	9,603	9,603
		# of test documents	6,746	3,680	3,662	3,299	3,299
		# of categories	135	93	92	90	10
System	**Type**	**Results reported by**					
Word	(non-learning)	[Yang 1999]	.150	.310	.290		
	probabilistic	[Dumais et al. 1998]				.752	.815
	probabilistic	[Joachims 1998]					.720
	probabilistic	[Lam et al. 1997]	.443 (MF_1)				
PropBayes	probabilistic	[Lewis 1992a]	.650				
Bim	probabilistic	[Li and Yamanishi 1999]				.747	
	probabilistic	[Li and Yamanishi 1999]				.773	
Nb	probabilistic	[Yang and Liu 1999]				.795	
	decision trees	[Dumais et al. 1998]					.884
C4.5	decision trees	[Joachims 1998]					.794
Ind	decision trees	[Lewis and Ringuette 1994]	.670				
Swap-1	decision rules	[Apté et al. 1994]		.805			
Ripper	decision rules	[Cohen and Singer 1999]	.683	.811		.820	
SleepingExperts	decision rules	[Cohen and Singer 1999]	**.753**	.759		.827	
Dl-Esc	decision rules	[Li and Yamanishi 1999]				.820	
Charade	decision rules	[Moulinier and Ganascia 1996]		.738			
Charade	decision rules	[Moulinier et al. 1996]		.783 (F_1)			
Llsf	regression	[Yang 1999]		.855	.810		
Llsf	regression	[Yang and Liu 1999]				.849	
BalancedWinnow	on-line linear	[Dagan et al. 1997]	.747 (M)	.833 (M)			
Widrow-Hoff	on-line linear	[Lam and Ho 1998]				.822	
Rocchio	batch linear	[Cohen and Singer 1999]	.660	.748		.776	
FindSim	batch linear	[Dumais et al. 1998]				.617	.646
Rocchio	batch linear	[Joachims 1998]					.799
Rocchio	batch linear	[Lam and Ho 1998]				.781	
Rocchio	batch linear	[Li and Yamanishi 1999]				.625	
Classi	neural network	[Ng et al. 1997]		.802			
Nnet	neural network	[Yang and Liu 1999]				.838	
	neural network	[Wiener et al. 1995]			**.820**		
Gis-W	example-based	[Lam and Ho 1998]				.860	
k-NN	example-based	[Joachims 1998]					.823
k-NN	example-based	[Lam and Ho 1998]				.820	
k-NN	example-based	[Yang 1999]	.690	.852	**.820**		
k-NN	example-based	[Yang and Liu 1999]				.856	
	SVM	[Dumais et al. 1998]				.870	**.920**
SvmLight	SVM	[Joachims 1998]					.864
SvmLight	SVM	[Li and Yamanishi 1999]				.841	
SvmLight	SVM	[Yang and Liu 1999]				.859	
AdaBoost.MH	committee	[Schapire and Singer 2000]		**.860**			
	committee	[Weiss et al. 1999]				**.878**	
	Bayesian net	[Dumais et al. 1998]				.800	.850
	Bayesian net	[Lam et al. 1997]	.542 (MF_1)				

Text mining: document clustering

- Motivation
  - Automatically group related documents based on their contents so that the documents in the same group are more similar than ones in other groups.
  - No predetermined training sets or taxonomies
  - Generate a taxonomy at runtime



Text mining: document clustering

- Clustering Process
  - Data preprocessing: remove stop words, stem, feature extraction, lexical analysis, etc.
  - Clustering methods:
    - **Hierarchical clustering**: compute similarities applying clustering algorithms.
    - K-means
    - Model-Based clustering (Neural Network Approach): clusters are represented by “exemplars”. (e.g.: SOM)

Hierarchical Document Clustering:HAC

- Hierarchic agglomerative clustering(HAC):distance-based hierarchic clustering

0. ***Input***: *D::={d_1,d_2,…d_n};*
1. Calculate similarity matrix SIM[i,j]
2. Repeat
3. Merge the most similar two clusters, K and L, to form a new cluster KL
4. Compute similarity (aggregation measure) between KL and each of the remaining cluster and update SIM[i,j]
5. Until there is a single (or specified number) cluster
6. ***Output:*** *dendrogram of clusters*

- Advantage:
  - producing better quality clusters
  - works relatively well in low dimension space
- Drawback:
  - distance computation in high dimension space
  - quadratic time complexity

Ref:[5][7][8][9][10][15][16][29]

Document Clustering:k-means

- k-means: distance-based flat clustering

0. ***Input***: *D::={d_1,d_2,…d_n}; k::=the cluster number;*
1. Select k document vectors as the initial centroids of k clusters
2. Repeat
3. Select one vector d in remaining documents
4. Compute similarities between d and k centroids
5. Put d in the closest cluster and recompute the centroid
6. Until the centroids don't change
7. ***Output:*** *k clusters of documents*

- Advantage:
  - linear time complexity
  - works relatively well in low dimension space
- Drawback:
  - distance computation in high dimension space
  - centroid vector may not well summarize the cluster documents
  - initial *k* clusters affect the quality of clusters

Ref:[5][7][8][9][10][15][16][29]

Text mining: document clustering

- Evaluation
  - Difficult : No predetermined training / test sets or taxonomies
  - Use a ground truth P'
    - Apply pair counting to compare the obtained partition P with the ground truth P' with external measures (Pair counting, Cluster matching, Information theory : Mutual information, V measure)
- Pair counting (Jaccard, Rand Index)

1- Count the pairs of elements classified into the same (different) clusters in the both partitions P and P':

a : number of pairs of elements classified together in P and P'
b : number of pairs of elements classified together in P and in different clusters in P'
c : number of pairs of elements classified in different clusters in P and together in P'
d : number of pairs of elements classified in different clusters in P and in P'

2- Built a confusion matrix

		Ground Truth	
		$C(v_i)=C(v_j)$	$C(v_i)\neq C(v_j)$
Clustering Result	$C(v_i)=C(v_j)$	a	b
	$C(v_i)\neq C(v_j)$	c	d

Text mining: document clustering

- Evaluation with pair counting

		Ground Truth	
		$C(v_i)=C(v_j)$	$C(v_i)\neq C(v_j)$
Clustering Result	$C(v_i)=C(v_j)$	a	b
	$C(v_i)\neq C(v_j)$	c	d

- Rand Index (Rand1971) : proportion of pairs classified in the same way in P and P' (same cluster or different cluster)

  Rand(P,P') = $\frac{a+d}{a+b+c+d}$
- Jaccard

$$Jaccard(P,P') = \frac{a}{a+b+c}$$

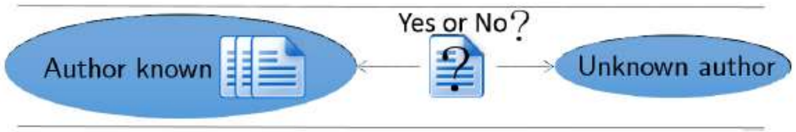
- values between 0 et 1 (concordance)
- Great variability of the index values on random partitions
- Extended versions
  - Normalized Rand Index(Hubert &Arabie1985, Meila2007)
  - Mirkin metric (Mirkin,1996)

Text mining: author identification

- A specific case of classification:

  Given a set of documents written by the same author, decide if a new document has been written by the same author as the others.

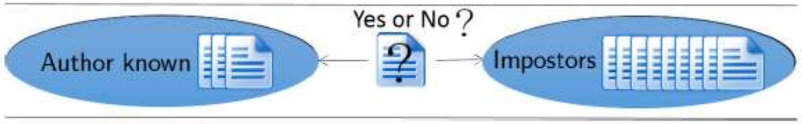
- Difficulty:
  - the set of known documents can be reduced to one element
  - Lack of negative example



Text mining: author identification

- Various strategies

  - Impostors by [Seidman2013]: add negative examples



How to find the impostors?

Text mining: author identification

- Frery, Largeron, Mathieu (ICDAR2015 + PANclef2014)

Three basic steps:

1. Pre processing

   ex. format to only lower-case letters

   ex. remove accents

2. Text representation (vectorize documents)

3. Two methods
   - Dissimilarity counter method
   - Decision tree + various representation spaces

Text mining: author identification

- Dissimilarity Counter method
  - Counts the number of documents d_i from A which are less dissimilar from u than from another document d_j of A
  - Output:
    - True for same Author if count > θ
    - False for differentAuthor if count < θ

  - Counter formula

Counter formulae

$$count(u_p) = |\{d_i \in A_p / min\{s(d_i, d_j), d_j \in A - d_i\} > s(d_i, u_p)\}|$$

Text mining: author identification

Dissimilarity Counter Method

Example

Let :

- $d_i \in A$ with $i \in \{1, .., |A|\}$ the known documents written by one author
- u the questioned document
- $\theta = \frac{|A|}{2}$

We compare documents using the cosine similarity. The counter is set to 0.

Dissimilarity matrix

	d1	d2	d3	d4	u
d1	0	0,78	0,77	0,79	0,71
d2	0,78	0	0,79	0,8	0,68
d3	0,77	0,79	0	0,8	0,78
d4	0,79	0,8	0,8	0	0,65
u	0,71	0,68	0,78	0,65	0

count = 0

Count =+1 since sim(d_1,u)< sim(d_1,d), d ∈ $A - \{d_1\}$

Text mining: author identification

Example

Let :

- $d_i \in A$ with $i \in \{1, .., |A|\}$ the known documents written by one author
- u the questioned document
- $\theta = \frac{|A|}{2}$

	d1	d2	d3	d4	u
d1	0	0,78	0,77	0,79	0,71
d2	0,78	0	0,79	0,8	**0,68**
d3	0,77	0,79	0	0,8	0,78
d4	0,79	0,8	0,8	0	0,65
u	0,71	0,68	0,78	0,65	0

count = 2

Text mining: author identification

Dissimilarity Counter Method

Example

Let :

- $d_i \in A$ with $i \in \{1, .., |A|\}$ the known documents written by one author
- u the questioned document
- $\theta = \frac{|A|}{2}$

	d1	d2	d3	d4	u
d1	0	0,78	0,77	0,79	0,71
d2	0,78	0	0,79	0,8	0,68
d3	0,77	0,79	0	0,8	0,78
d4	0,79	0,8	0,8	0	0,65
u	0,71	0,68	0,78	0,65	0

count = 2

Text mining: author identification

Dissimilarity Counter Method

Example

Let :

- $d_i \in A$ with $i \in \{1, .., |A|\}$ the known documents written by one author
- u the questioned document
- $\theta = \frac{|A|}{2}$

	d1	d2	d3	d4	u
d1	0	0,78	0,77	0,79	0,71
d2	0,78	0	0,79	0,8	0,68
d3	0,77	0,79	0	0,8	0,78
d4	0,79	0,8	0,8	0	0,65
u	0,71	0,68	0,78	0,65	0

count = 3, $count > \theta$ –> sameAuthor

Text mining: author identification

- Dissimilarity Counter method
  - Limits
    - Handles only one representation space
    - Can not treat problems with only one document in A
- Decision tree with several representation spaces

Representation space	
Term	Model
Character 8-grams	tf-idf
Character 3-grams	tf-idf
Word 2-grams	tf-idf
Word 1-gram	tf-idf without the 30% most frequent words
Word 1-gram	tf-idf without stop words
Phrases	word per sentence mean and std dev
Vocabulary diversity	total number of different terms divided by the total number of occurrences of words
Punctuation	average of punctuation marks per sentence characters : "," ";" ":" "(" ")" "!" "?"

Text mining: author identification

Building our decision tree classier:

- Learning step on a training set composed of different problems described by:

  - attributes representing the similarity between the known and the unknown documents for the given representation spaces

  - the true label of the problem (True or False)

- Evaluating on a test set by comparing label obtained with our classifier and the real ones

Text mining: author identification

- Attributes defined for eor each problem (A_p, u_p, Label(u_p)) and each representation space

Mean

$$mean(u_p) = \frac{1}{|A_p|} \times \sum_{d_i \in A_p} s(d_i, u_p) \quad (1)$$

Count

$$count(u_p) = |\{di \in A_p / min\{s(d_i, d_j), d_j \in A - d_i\} > s(d_i, u_p)\}| \quad (2)$$

TOT_{count}

$$TOT_{count}(u_p) = \sum_{v=1}^{8} count_v(u_p) \quad (3)$$

Text mining: author identification

Experimentation was made on the training corpora given by PAN-AI 14:

- Four different languages : Engligh, Dutch, Spanish and Greek
- Four different types : Article, Novel, Review and Essay

Corpus	EN	EE	DR	DE	SP	GR
Problems#	100	200	100	96	100	100
AUC	89%	70%	68%	91%	77%	76%

Table: 10-fold cross validation on the training corpus

Corpus	EN	EE	DR	DE	SP	GR
Time(min)	3:10	0:54	0:08	0:29	1:00	0:57
Final rank	3/13	3/13	3/13	4/13	3/13	3/12

Table: Challenge evaluation execution time results

Document visualization

- Cloud of words
- Representation of terms and documents in reduced space
  - Word Embedding
  - LSA - Correspondance analysis on contingency table (Documents x terms)

Outline

- Introduction
- Preprocessing
- Features extraction - indexing
- Weighting models
- Features Extraction - dimension reduction
- Text mining (association - categorization – clustering)
- References

- Application with R

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